

**AUTOMATED PART TRACEABILITY AND DATA
SYNCHRONIZATION**

EMD51807 - PROJECT REPORT

Submitted by

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UNDER THE GUIDANCE

OF

Dr. S. JANAKIRAMAN

In partial fulfilment for the award

of

BACHELOR OF TECHNOLOGY

in

MECHATRONICS



HINDUSTAN
INSTITUTE OF TECHNOLOGY & SCIENCE
(DEEMED TO BE UNIVERSITY)

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APRIL 2026



BONAFIDE CERTIFICATE

This is to certify that this project report **“AUTOMATED PART TRACEABILITY AND DATA SYNCHRONIZATION”** is the Bonafide work of **PAUL SAMUEL D (22125003), BALAMURUGAN B (22125013), THIRISHA R (22125022)** who carried out this project work as a part of the **EMD51807 PROJECT** under my supervision during the 8th Semester of the academic year 2025-202.

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During the internship period, **he** demonstrated a strong willingness to learn, professionalism, and dedication towards assigned responsibilities. **His** contributions were valuable, and **he** consistently met expectations through sincere efforts and a positive attitude.

We found **him** to be hardworking, disciplined, and proactive. We wish **him** every success in **his** future academic and professional endeavors.

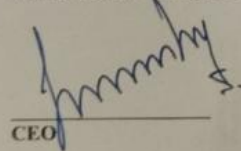
Project Worked: Automated Part transferability & Data Synchronization

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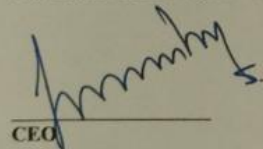
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During the internship period, **she** demonstrated a strong willingness to learn, professionalism, and dedication towards assigned responsibilities. **Her** contributions were valuable, and **she** consistently met expectations through sincere efforts and a positive attitude.

We found **her** to be hardworking, disciplined, and proactive. We wish **her** every success in **her** future academic and professional endeavors.

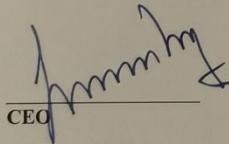
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DEPARTMENT OF MECHATRONICS ENGINEERING

PROJECT PHASE 2 (EMD51807)

Project Batch No : 13

Title of The Project : **AUTOMATED PART TRACEABILITY
AND DATA SYNCHRONIZATION**

Number of The Students In The Batch : 3

Student's Individual Contribution

Name of the Student: PAUL SAMUEL D

Register No.:22125003

Contribution: Conducted literature review, supported both hardware and software development, and contributed to system analysis and documentation.

I was responsible for reviewing research papers related to conveyor systems, ERP integration, barcode systems, and object detection models. I identified key concepts and research gaps, which helped shape the project design. I also assisted in both hardware and software development, testing, and result analysis, ensuring that the system performance aligns with project objectives.

PAUL SAMUEL D (22125003)



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PROJECT PHASE 2 (EMD51807)

Project Batch No : 13

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AND DATA SYNCHRONIZATION**

Number of The Students In The Batch : 3

Student's Individual Contribution

Name of the Student: BALAMURUGAN B

Register No.:22125013

Contribution: Handled complete hardware integration including conveyor system design, motor setup, sensor integration, and mechanical assembly.

I was responsible for building and integrating all hardware components of the system. This includes designing the conveyor structure using aluminum profiles, assembling pulleys, belts, bearings, and motor systems, and ensuring proper alignment for smooth operation. I also worked on integrating the Raspberry Pi, camera, motor driver, and power supply, making sure all components function together reliably.

BALAMURUGAN B (22125013)



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PROJECT PHASE 2 (EMD51807)

Project Batch No : 13

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AND DATA SYNCHRONIZATION**

Number of The Students In The Batch : 3

Student's Individual Contribution

Name of the Student: Thirisha. R

Register No.:22125022

Contribution: Developed the complete software system including YOLOv8-based object detection, ERP dashboard, barcode generation, and real-time data synchronization.

I was responsible for designing and implementing the full software architecture of the APTDSC system. This includes training the YOLOv8 model for detecting nuts and bolts, developing the ERP interface using HTML and JSON, and integrating barcode generation for traceability. I also worked on real-time data processing, API communication, and system logic to ensure smooth interaction between detection, storage, and visualization modules.

THIRISHA.R (22125022)

ABSTRACT

The manufacturing industry is transforming at a rapid pace as manufacturers adopt Industry 4.0 technologies, including implementing smart devices and utilizing real-time information. One of the major issues confronting production lines today is the reliance on manual labeling for parts. Manual labeling introduces human error in the movement of parts, which leads to a lack of traceability. These two major problems will delay production, require rework and reduce the overall efficiency of the manufacturing process.

The Automated Part Traceability and Data Synchronization (APTDS) solution has been designed to address this issue; it incorporates a combination of computer vision, motion control and ERP integration. The general structure of the APTDS employs a Raspberry Pi 5 as the primary processor to identify and classify industrial fasteners. These fasteners include M5, M6, M8 and M10 nuts and bolts. The YOLOv8 model is utilized to identify and classify the fasteners utilizing the APTDS. APTDS has demonstrated approximately 87% accuracy through real-time processing capabilities and thus readily lends itself to use on a moving conveyor system.

After a part has been identified and validated, a unique part label is produced automatically by the system, and the part information is synchronized with an ERP system in real time. The ISO-compliant barcode that is created is interoperable, trustworthy, and consistent throughout manufacturing environments and enables validation-based workflows that allow only validated and authorized parts to be processed further along the production line.

The new system completed the automatic identification and validation of parts thereby reducing operator error, improving part traceability, and connecting shop-floor activities to their corresponding digital counterpart. All in all, APTDS is a cost-effective and scalable option for supporting the improved efficiency, standardization, and digital traceability of current manufacturing operations.

Keywords: Industry 4.0, Data Synchronization, Computer Vision, ERP Integration, ISO standard barcode generation, automated conveyor systems, real-time monitor.

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CHAPTER-1

INTRODUCTION

The advent of the Fourth Industrial Revolution has altered the way that traditional manufacturing operates and has shifted the focus of manufacturing to an increasingly intelligent, interconnected, and data-centric ecosystem. Today's production systems are dependent upon advanced automation, real-time monitoring, and seamless integration of physical processes with digital networks, rather than relying on mechanical operations. Consequently, accurate part traceability and synchronization between shop-floor operations and enterprise systems have become vital priorities.

Nonetheless, too many companies currently utilize inaccurate outdated manual methods for part labeling and tracking which are prone to a variety of human error. Any inaccuracy in either part identification, labeling, or tracking can cause incorrect parts to be introduced into the production line, causing considerable quality issues, excessive amounts of rework, and delays in production. In addition, the inability to synchronize production data in real-time with ERP systems will further impede the ability to provide consistent and reliable manufacturing data. An integrated solution known as the Automated Part Traceability and Data Synchronization (APTDS) system is proposed to address these issues. The APTDS System will utilize the combined capabilities of computer vision, embedded systems, and connectivity to ERP systems to enable automation of the parts detection, validation, labeling, and data synchronization processes. By implementing intelligent validation mechanisms and facilitating the real-time exchange of data, only verified parts will be allowed to flow through the production process.

In addition to increasing traceability, the APTDS helps to provide consistency and reliability to the processes. It serves as a connector for the physical manufacturing process with digital enterprise systems that enable the creation of efficient, non-defective, and scalable manufacturing workflows.

1.1 OBJECTIVES OF THE PROJECT

Automated Part Traceability and Data Synchronization (APTDS) is the primary purpose of this project is to create an intelligent and dependable system to automate the identification, labeling and tracking of parts used in a manufacturing environment. The

APTDS project is to reduce human interaction in routine tasks which reduces errors that typically occur with manual labeling systems. Utilizing computer vision technology will enable the system to identify and classify industrial parts such as M5, M6, M8 and M10 nuts and bolts, etc. Once parts are identified, the system will create a unique serialized label and barcode (following ISO standards), therefore maintaining consistency and compatibility across all industrial applications. Another purpose of the APTDS system is to achieve real-time synchronization of data between the shop floor and ERP systems providing seamless tracking and monitoring of parts throughout the production process. Furthermore, validation will be included in the APTDS system to allow only validated parts to advance in the production process. In conclusion, the APTDS project will improve traceability, productivity and reduce errors/delays in production.

1.2 SCOPE OF THE PROJECT

This project includes the design and creation of an automated system that provides a way to detect, validate, and synchronize data for parts in a controlled industrial environment. This system uses a conveyor mechanism to move parts in conjunction with various sensors and motor controls to assist with regulating the flow of parts. The core processing unit, Raspberry Pi 5, is used to run the YOLOv8-based computer vision model for real-time detection and classification of selected industrial parts. It also includes automatic serialization and the creation of ISO standard barcodes, as well as the ability to synchronize them with an ERP system, allowing for real-time data processing. One major aspect of this system is that it can validate a part before granting it further access, ensuring greater accuracy and reliability throughout a workflow. All of these systems are currently implemented at a prototype level only for specific component sets and do not currently support large-scale production environments or high-level ERP integrations. However, this project is a nice base for future expansion and for eventual real-world uses.

CHAPTER-2

LITERATURE REVIEW

With modern manufacturing systems being used in Industry 4.0, the manufacturing industry is changing from manual manufacturing processes to automated, intelligent and connected manufacturing environments. Within today's high-paced, competitive industrial environment; efficiency, precision and traceability are necessary to ensure quality and minimize total production costs. Some of the technologies that have enabled this redesign are industry 4.0 technologies such as IoT (Internet of Things), Enterprise Resource Planning, computer vision and automated material handling.

A major difficulty that many industries have is accurately identifying and tracking parts during all stages of production. Tracking parts with manual labels and tracking manually provides a substantial level of human error resulting in improperly handled parts, delays in production, increased rework and unaccountability due to insufficiently maintained records of tracking. The inability for shop floor and enterprise level operations to be synchronized in real time introduces gaps in the real time visibility of data, making it very difficult for manufacturers to maintain a consistent and efficient work flow in their factories.

To overcome these challenges a number of technologies have been studied with respect to their viability; these include; conveyor automation, vision inspection systems, ERP systems and barcode identification. The conveyor system is the primary method in which material is handled in an automated fashion allowing for consistent and smooth flow of components through each of the manufacturing processes. The belt mechanism on the conveyor provides smooth and stable movement, which helps to ensure accuracy in the inspection and processing of products. Similarly, the ERP system provides an effective and efficient means of creating a centralized repository for all production data and will also help to enhance decision-making and process control. Because they are dependable, standardized, and simple to integrate with digital technologies, bar code systems (especially those that comply with ISO specifications) have become the preferred method of part identification and tracking. The efficiency of substantial part-processing operations, using conveyor-based and automated tracking systems, can be assessed through worldwide implementations, such as bag handling at

airports; by automating the tracking of large quantities of parts, very slight margin of error can be expected. These implementations emphasize the need for intelligent tracking and verification capabilities, and additional automation to be used in conjunction with them. Even though considerable advancement has occurred, a vast majority of existing systems are not fully integrated and work in isolation from one another; consequently, this creates a problem where there is no seamless connection between detection, validation, and data synchronization functions. Thus, a unified comprehensive solution must be developed that links together the physical operation site and the digital operating system. The Automated Part Traceability and Data Syncing (APTDS) will ultimately fill in this gap between digital and physical processing by providing an integrated system that uses; (1) computer vision; (2) Automated conveyor systems; (3) the creation of a barcode; and (4) the synchronization of ERP systems.

The purpose of this chapter will be to provide an extensive literature overview regarding; (1) the existing literature pertaining to varying types of conveyors; (2) the integration of ERP's; (3) ISO bar code systems; and (4) other technologies related to belt systems and conveyor systems at airports in order to (a) review the present methods being utilized; (b) identify limitations within those methods; and (c) gain better insight into the foundation necessary for the proposed system to provide accurate, real-time, and fully automated traceability of parts in today's manufacturing environment.

2.1 LITERATURE REVIEW AND INFERENCE

Gupta K. et al. [1] studied automated handling systems, focusing on conveyor motion and positioning control. This study shows that incorrect positioning/movement of conveyor can lead to inspection mistakes, requiring that there is a consistent speed and distance between each item being moved so that inspection can be performed correctly.

Inference: The importance of consistently controlled conveyor movement is a major finding of this study, illustrating how the level of stability affects the performance of equipment used for detection of items. Since imaging in APTDS will require controlled movement with the conveyor, it is necessary to consider how to design conveyors to provide reliable imaging capabilities.

Zhang L. et al. [2] developed a smart conveyor inspection system using PLC and sensor technology that improves the positioning of items during inspection. This system uses sensors to detect if an item is on the conveyor and the timing of the item to synchronize the inspection process with the movement of the item.

Inference: The control of items on the conveyor by the sensor provides the consistency needed to complete the inspection process. This consistency also applies to the validation of APTDS imaging systems. The sensor-based control of the automated inspection process reduces the number of humans needed to complete the inspections, thus providing greater accuracy and efficiency.

P. Verma et al. [3] (2019) studied the implementation of ERP systems in small scale manufacturing companies. Their study shows that using ERP systems removes the risk of entering incorrect information into or out of an organization's database. Also, it improves communications between departments such as production and inventory.

Inference : ERP systems help eliminate human error and improve workflow efficiency. They provide a centralized location for storing data. This allows organizations to track their information in real time using APTDS. Additionally, ERP systems provide transparency to a company's manufacturing processes. Therefore, the integration of ERP systems into an organization is essential to automate its processes.

R. Kumar et al. [4] (2020) conceptualized the integration of ERP into a traceability system that allows for real-time synchronization between shop-floor data and enterprise data. This increases the transparency and consistency of the two types of data. The authors also stated that the real-time synchronization of data will decrease the number of delays that occur due to manual data handling.

Inference : Real-time synchronization enhances traceability and accountability. It also improves the overall consistency of data. The concept of real-time synchronization can be applied directly to APTDS and will help maintain accurate product records. Therefore, the integration of ERP systems into organizations is critical.

M. Singh et al. (2020) focuses on improving predictive maintenance through sensor-based monitoring of performance/early fault detection. Results show the use of predictive maintenance improves both system downtime and efficiency. In addition,

the IoT-based conveyor system will permit remote monitoring of conveyor systems and is consistent with Industry 4.0 practices.

Inference: IoT will increase reliability through advanced integration techniques while providing predictive capabilities for when a system may fail. This will result in greater operational efficiencies than would otherwise be realized without the help of IoT solutions. Likewise, the use of similar IoT technology would enhance robustness in APTDS, thereby providing additional benefits through smart monitoring of equipment.

A. Singh et al. (2020) facilitates seamless communication between machines and databases for data exchange via real-time updates versus waiting to receive and send data across systems.

Inference: Using APIs for ERP communication allows for real-time synchronization of data between systems resulting in loss of data (receiving prior to sending, etc.) , thus providing direct efficiencies into automating APTDS operations through intelligent software applications.

A new vision-based component detection system developed by J. Lee et al. [7] (2021), utilized cameras and image processing algorithms to classify parts and has demonstrated a high precision in detecting similar components. Effective and appropriate lighting and positioning were the noted contributors to the high accuracy in their testing.

Hence, the inference can be made that: By incorporating vision systems, detection accuracy has been significantly improved, thus allowing for the automation of inspection processes. This supports APTDS using YOLO as their detection method.

L. Wang et al. [8] (2021), developed a conveyor monitoring system using Internet of Things (IoT) technology to identify faults such as misalignment & motor issues, provide real-time alerts for maintenance, improve overall safety and reliability, and reduce the number of unexpected failures.

Inference : Real-time monitoring of the systems has improved overall system stability through early fault detection.

S. Mehta etc.[9] (2021) created a smart ERP for Industry 4.0 that allows for real-time analytics and decisions. In doing so, there is increased transparency of production and

operational efficiency. Furthermore, data is used differently - via Data Driven Management.

Inference: The analytics from ERP influence the visibility and control provided to the company, providing improved decision making, which aligns with the objectives of the APTDS Project. The use of ERP improves traceability within a work process, so ERP is important.

H. Brown et al. [10] (2021) presented important information about how barcodes provide interoperability and standardization between devices, as well as consistent encoding and decoding of useable data. Barcodes are profoundly common to most industries today, whether used to track components in manufacturing environments or to ensure that parts arrive at their proper destination.

Inference: The use of ISO barcodes provides consistency and accuracy of labeling of parts. This consistency supports accurate tracking and is directly applicable to APTDS. The application of standardization enhances system interoperability, and therefore, barcode systems are important.

D. Roy et al. [11] (2022) developed cloud-based ERP systems to support Manufacturing. The implementation or the development of these systems supports remote access and scalable handling of data; therefore, they support centralized storage of data.

Inference: By utilizing Cloud ERP options, companies can increase their capacity for scalability and accessibility to support future growth, which is essential for APTDS to develop, and it allows for flexibility within the overall design of the system; thereby, integration of the Cloud option is desirable.

A. Sharma et al. [12] (2022) Automated conveyor systems with synchronized control develop the performance benefits; by lowering the need for manual labour and improving efficiency. It provides a means of facilitating the movement of the product and inspection processes.

Inferences made from these two articles conclude that increasing the synchronization between processes helps to create accurate detection of events; by producing an accurate timing between events. This has a positive impact on how well APTDS will

function. It is important to be able to move the product in a controlled manner. Therefore, synchronization is important.

K. Patel et al. [13] (2022) the authors of the second article believe that the introduction of ISO barcode generation rates for accurate labelling/tracking will reduce the number of human errors in the process.

Inferences made from these two articles show that automated labelling will create accurate quantities and help provide consistency in the tracking of the product. This helps support the design of APTDS. Furthermore, APTDS reduces the number of manual errors. Therefore, barcode automation is vital to the success of APTDS.

N. Das and colleagues (2022) have developed a conveyor mechanism that uses belt drives as the means of moving material through industry more efficiently. This approach addresses the need to create improved contact (or grip) between the belt and other components to reduce slippage while in motion.

Inference: This research demonstrates the importance of belt stability for providing consistent motion. Stable transport increases both inspection system performance and accuracy. In Automatic Parts Detection Systems (APDS), this same principle is followed to provide reliably positioned parts for inspection.

S. Rao et al. (2022) have presented an Automated Tracking System (ATS), which combines bar code scanning with Enterprise Resource Planning (ERP) systems as a method of efficiently tracking industrially produced goods. By automatically capturing data from the materials placed on the conveyor, then instantaneously synchronizing that data with a central database.

Inference: The integration of bar code scanning systems with ERP provides real-time visibility of manufacturing data. As a result, accurate tracking of materials and components occurs from start to finish. The same concept is applicable to Automatic Parts Detection Systems (APDS) for matching the detected parts with their corresponding entries in the ERP.

M. Chen et al [16] (2023) studied automated airport baggage handling systems that use conveyor networks for sorting and routing the items. The system can process a large

number of items quickly and with little to no human assistance. This system uses automated tracking and routing to ensure proper delivery of all items.

Inference: Speed control of movement is a key factor for maintaining operational efficiency of the system; Without proper speed of movement there will be a greater chance that inspected items are not detected and/or properly timed between detection/inspection processes.

T. Nguyen et al [18] (2023) developed an AI defect detection inspection system integrated with conveyor belts which provides for real time defect detection of faulty products. This system uses machine learning algorithms to accurately identify defects. The system was able to operate successfully under dynamic industrial environments.

Inference: AI Systems have improved the overall accuracy and confidence of defect detection. AI Systems allow for real time decisions to be made within an automated industrial work environment. In APTDS, YOLO detection works in a similar manner.

J. Park et al. (2023) have created an ERP–MES integrated system to link the operations of the shop floor and management from Enterprise level to facilitate improved and timely coordination between the production and planning processes through the seamless exchange of information at real time.

Inference: The integration of ERP and MES systems leads to a more coordinated workflow in the overall production process by supplying real-time synchronization of production-related information.

A. Khan et al. (2023) analysed the automated conveyor sorting systems utilized in airport operations to efficiently handle large amounts of baggage as quickly as possible. The sorting system provides a means for accurately routing the items being delivered to their destinations..

Inference: Automated sorting systems provide an improvement in accuracy and decrease the amount of human error that occurs as a result of these systems. Therefore, they guarantee that there are no components that cannot be correctly routed within a dynamic environment.

R. Patel et al. [22] (2024) found that using free-flowing machine learning to maintain a conveyor system results in lower costs associated with delayed repairs, as well as

making it possible to maintain the system. Thus, spending less money on a machine that breaks down and decreases its usefulness allows the machine to operate longer; consequently, it improves the performance of each conveyor within the conveyor system.

Inference: fewer unanticipated system failures and negatively impacted performance, predictive maintenance continues to be an effective method of ensuring that systems are managed proactively rather than reactively. This means there are more opportunities for the continued operation of systems and the possibility of offering better performance in the future through the use of the latest technology.

S. Verma et al. [23] (2024) developed an artificial intelligence ERP solution for manufacturing industries that utilizes data analytics to facilitate smarter decision-making. Their AI ERP solution automates various management processes that will improve efficiency. Additionally, the study indicates that use of data analytics improves both accuracy of planning and execution and supports intelligent workflow automation within an organization.

Conclusion: Artificial intelligence will enhance ERP functionality by facilitating smarter decisions through greater automation and intelligent operation of the system. Furthermore, the implementation of such advancements into an APTDS will allow for improved functionality of the ERP system by enabling better utilization of data and better control of processes. Therefore, the benefit of integrating an artificial intelligence-based solution into APTDS is an achievable benefit.

L. Zhao et al. [24] (2024) produced a high-speed conveyor designed for the optimization of using automated systems for industrial processes. The development of high-speed conveyors will enable continual and efficient movement of materials through a facility. Additionally, the production of high-speed conveyors will provide a stable foundation for high-speed motion. The study stresses that synchronization is critical to the successful operation of high-speed systems. This finding may contribute to the development of a more productive manufacturing system through greater levels of productivity.

Conclusion: To successfully operate a high-speed system, both precision control and synchronization are necessary for successful operation. If precision control is not

performed as required, the likelihood of increased errors will occur. Within APTDS, maintaining a controlled speed will allow for precision detection using automated equipment. Hence, synchronization is essential for successful operation of high-speed systems.

T. Puglisi et al. [25] (2022) presents an overview of intelligent automation, with an emphasis on adaptive learning, multimodal sensing, and real-time evaluation frameworks. Additionally, the importance of flexibility, scalability, and adaptability in developing future-oriented intelligent automated systems is discussed as well as challenges associated with lack of standardization in automated processes.

Inference: current trends in developing intelligent automation systems have moved toward designing these systems as adaptive and intelligent designs. APTDS has similarly applied the principles of adaptive design through the use of integrated detection and synchronization components as part of an overall approach that integrates multiple technologies into a unified solution. Therefore, adaptive design will continue to be an important component of any systems developed to meet future needs.

The literature shows that automated conveyor systems, Enterprise Resource Planning System (ERP) integration, barcode technology, and intelligent inspection systems have made major advancements in their respective fields. Automated conveyor systems have evolved from simple mechanical methods of moving materials to more advanced methods of moving materials. ERP systems have improved the ability to coordinate activities on the shop floor and manage businesses as a whole. This is accomplished through a higher degree of synchronization between real-time data coming from the shop floor and that being used by the businesses via the ERP system. The development of barcode technology has greatly increased accuracy in identifying parts and tracking their location, especially when using standard coding methods. Use of automated inspection systems has enhanced the ability to detect defects and reduced reliance on manual inspection. In addition to these studies, large-scale installations such as those implemented at airports, serve as examples of the effectiveness and reliability of automated tracking and sorting systems. Overall, the findings of the studies reviewed suggest that the integration of mechanical systems with digital technologies is necessary in order to create an efficient, accurate, and scalable manufacturing solution.

Research Gap

Despite all the progress that has been made over the last several years, there is still a lot of work to do. Most of the current solutions include only one element of an overall solution - such as a conveyor system, ERP platform, or bar code technology - but none provide complete solutions with full integration for all elements. For example, there is insufficient co-operation between part identification, part validation, part labelling, and part tracking using real-time data in the same system. In addition, a significant number of systems do not provide any intelligent validation methods to prevent a part from continuing through the production process that is not correct. There are significant limitations to the way edge AI models can detect parts in real time in low resource environments. Furthermore, computer vision manufacturing systems and ERP systems are not typically integrated well in a real-time application.

To resolve these concerns, an integrated solution called the Automated Part Tracking and Distribution System (APTDS) combined all of the elements mentioned above and will provide you with a comprehensive traceability solution for modern manufacturing operations.

CHAPTER-3

METHODOLOGY

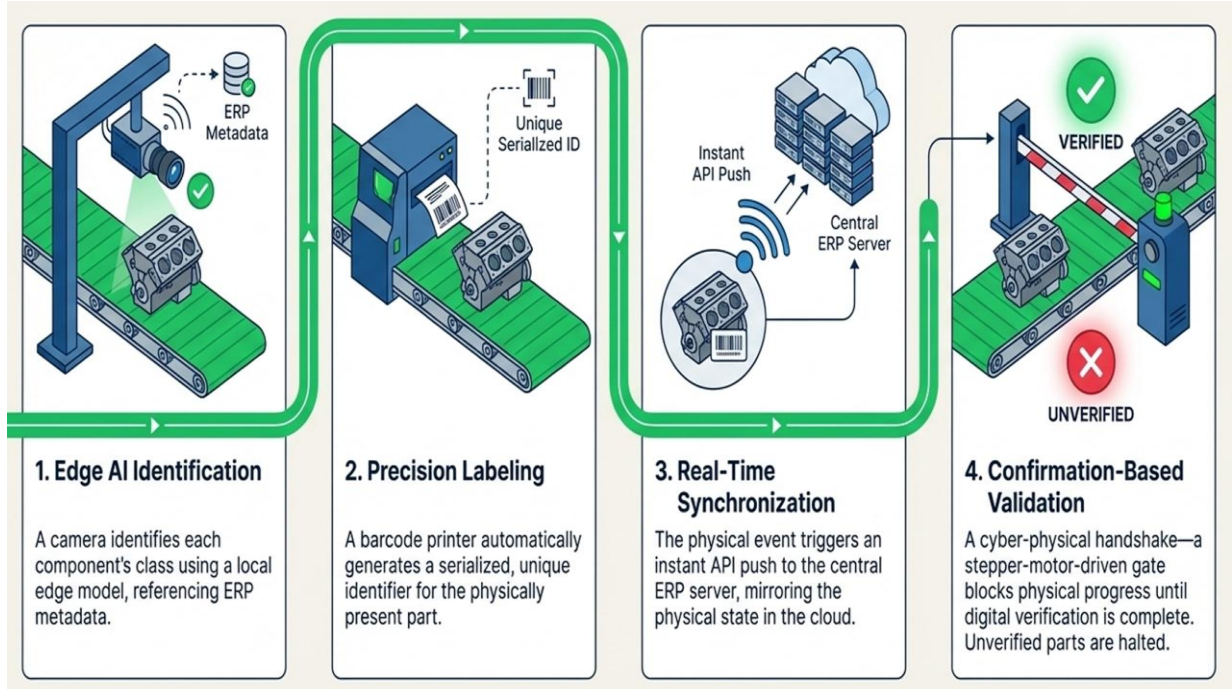


Figure 1: Workflow of Automated Part Traceability and Data Synchronization (APTDS)

The working flow of the Automated Parts Traceability and Data System (APTDS) is illustrated in Figure 1. The process illustrates how the system identifies, labels, synchronises and validates physical objects within an automated manufacturing environment. As depicted in the diagram, the integration of Computer Vision, Labelling Systems, ERP Connectivity, and Validation Mechanisms is streamlined into one continuous process.

The APTDS consists of multiple components; each of these components plays a role in the completion of the entire working flow. The beginning of this working flow starts with Edge AI Identification, which consists of a camera that has been installed above the conveyor line that will capture images of the components as they move down the conveyor system. Once images are captured, a locally deployed Edge AI model will then identify the part type (i.e., nuts, bolts, etc.) and part size (i.e., small, medium and/or large) based on the captured images of the moving parts. The identification process occurs in real time at the edge device (in this case, the Raspberry Pi). The

APTDS does not rely upon external servers to accomplish this identification process, thereby reducing latency and providing a fast decision-making process. In addition to identifying and validating the moving part as nut or bolt based upon dimensions, the APTDS will use the ERP metadata associated with each moving part that was captured on the conveyor to ensure the accuracy of the identification process from the beginning of the working flow.

After determining the component part, the system moves into the phase of Precision Labelling. At this stage, a barcode printer will generate a unique serialized ID for each identified component, following ISO specifications so that it can be read throughout the world by all industrial systems. The labelling is entirely automated with no human involvement, therefore reducing the potential for human error. By giving each physical component a unique identifier, the physical component(s) will also have a digital relationship with each other and create a basis for traceability.

After the labelling stage, the system will move on to Real-Time Synchronisation, which links the physical process and digital ERP system. Once an item has been labelled, the data related to that item is instantly transmitted to a central ERP server using an API. As such, each productive action on the shop floor is immediately reflected within the digital system. The central digital ERP system will maintain records of part type, batch number, timestamp, and status for each part, which will allow for real-time monitoring and transparency of data across the manufacturing process.

Last but certainly not least, the fourth and final key element of the Cyber-Physical Systems (CPS) discussed above is Confirmation Based Validation (CBV) which is one of the core control functions within this system. The CBV is used as a gate system to validate the proper movement through the production line of identified and labelled components that are to be deemed accepted or not accepted based on the required specifications in the company's ERP system (Enterprise Resource Planning System). To do this, the stepper motor gate uses confirmation from the company's ERP system to validate the identified component against the established specification. If it does, then the gate allows the component to continue to the next step of production or if it does not, the gate blocks the identified and labelled component from continuing through the production line. Thus, CBV limits the movement of only compliant and

validated components through the production line, thereby minimizing production errors and assuring the quality of products being produced.

In general, as illustrated in Figure 1 above, the entire cyber-physical system operates as a fully integrated physical and digital validation process by providing greater accuracy and traceability of objects, combined with an automated and reliable decision making process to minimize human intervention and improve industrial efficiencies.

3.1 BLOCK DIAGRAM

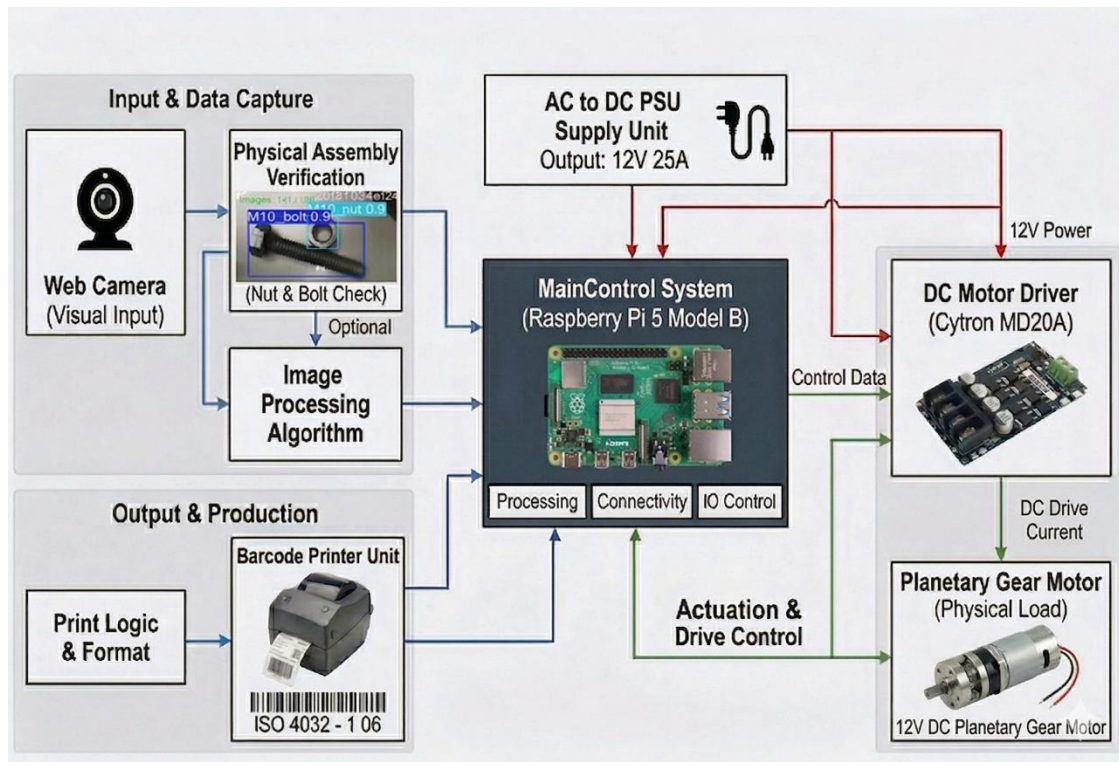


Figure 2: Block Diagram of APTDS System Architecture

The overall system architecture of APTDS (Automated Part Traceability and Data Synchronization), illustrating the various hardware and software components that work together to automatically detect, label and control parts is outlined in Figure 2. The process starts off with a web camera continuously capturing images of the components that are placed onto the conveyor in an input/data capture stage. The photos that are captured by the web camera are sent to an image processing module, which is running on a Raspberry Pi 5, so that image processing can be performed in real-time via the YOLO (You Only Look Once) v8 object detection/classification algorithm in order to scan the parts for classification as brackets/nuts/bolts etc. The YOLO v8 algorithm will

quickly and accurately scan visual data at the edge thereby allowing for fast and accurate identification of parts without needing to use any external server. A physical assembly verification step that is optional provides additional assurance that the part has been correctly characterized prior to continuing on with production.

Following the processing of the data obtained from all the sensors, it is sent to the main control unit (Raspberry Pi 5 Model B) to be evaluated and controlled as one main unit. The main control unit is responsible for computing data, communicating with other modules and making decisions regarding the operation of each unit depending on the type of part which was detected. During the output/production stages after the part was created, a unique serialized label/serial number will be generated by a barcode printer for each part that has been identified. The barcode will comply with the ISO standard; therefore it will provide for a universal method for reading barcodes, standardization of barcodes, and use of barcodes within the industrial world which will provide a reliable method for tracking and tracing of individual components during the entire production cycle.

This block diagram is an illustration of an integrated cyber-physical system that integrates a vision-based detection mechanism (YOLOv8) using standardised barcodes and labelling techniques, with real-time data processing and motor control as an operational and functional part of the system. The architecture of the system provides for direct decision-making, accurate part identification and the ability to automate reliably. As such, the Level of automation achieved in industrial settings is improved significantly with reduced need for manual intervention, resulting in increased overall productivity.

3.2 COMPONENTS USED



Figure 3: Web camera

The web camera used for the APTDS is a 12-megapixel USB camera that represents the primary source of visual input for detecting parts (see Figure 3: Web Camera). The camera allows for high resolution images of the parts travelling on the conveyor and therefore provides an opportunity for the system to accurately detect and identify the many small industrial parts such as nuts and bolts. Because the images are of a higher resolution, it will be possible to detect finer details (i.e. shape, edge and orientation) with a higher degree of accuracy than with lower resolution images. Further to that, the camera continuously transmits images of the parts in real-time so that the system can then use the images to identify and classify the parts as they pass by.



Figure 4: Raspberry pi 5

The APTDS System's central control unit is represented by the Raspberry Pi 5 Model B, which is pictured in Figure 4 (Raspberry Pi 5). The Raspberry Pi processes the input from the web camera, runs the YOLOv8 object detection algorithm to perform real-time object detection, and uses the information received to make decisions regarding the items detected. In addition, the Raspberry Pi communicates with the barcode printer, ERP system, and motor driver via a variety of interfaces (USB and GPIO), providing efficient control of all system operations due to its high processing capabilities, compact design, and low power consumption. As a result of its ability to enable a fully functional automation APTDS system with edge-based automation, the Raspberry Pi will coordinate all systems and processes.



Figure 5: DC Planetary Gear Motor

Another type of motor used in the APTDS System is the 12V DC planetary gear motor, which can be seen in Figure 5 (DC) Planetary Gear Motor. The planetary gear motor will provide controlled mechanical movement to the conveyor within the APTDS System, converting electricity to mechanical motion. The attached planetary gearbox provides greater torque and allows for smooth and controlled speeds, providing stable and accurate movement of the various parts being detected and validated. The DC motor driver is used to control the motor's speed and direction by receiving signals from the Raspberry Pi and allowing the motor to execute functions, including starting, stopping, and controlling the speed of the motor. Overall, the planetary gear motor provides efficient, reliable, and accurate transportation of all components within the APTDS System.

Table 1: Specifications of 12V DC Planetary Gear Motor

Parameter	Specification
Rated Voltage	12 V DC
Rated Current	1.5 – 3 A (typical load)
No Load Current	0.3 – 0.6 A
Stall Current	5 – 10 A
Rated Torque	10 – 25 kg·cm
Maximum Torque	30 – 50 kg·cm
Speed (No Load)	100 – 300 RPM
Gear Type	Planetary Gearbox
Efficiency	High

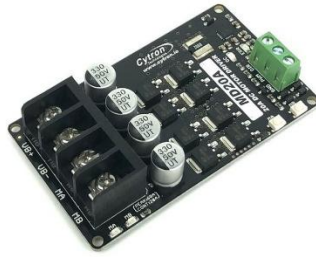


Figure 6: Cytron MD20A Motor Driver

The MD20A DC motor driver manufactured by Cytron Electronics as shown in exhibit 6 below is the component of the APTDS system that controls the speed and direction of movement of the primary motor of the APTDS system, a 12V DC planetary gear motor. The DC motor driver provides a method for the Raspberry Pi to communicate with the DC motor: accept commands from the Raspberry Pi and regulate the power provided to the motor.

Table 2: Specifications of Cytron MD20A Motor Driver

Parameter	Specification
Operating Voltage	6V – 30V DC
Continuous Current	20 A
Peak Current	60 A
Control Method	PWM & Direction Control
Logic Input Voltage	3.3V / 5V Compatible
Protection Features	Overcurrent, Overheat
Motor Type Supported	DC Brushed Motor
Interface	GPIO / PWM
Efficiency	High



Figure 7: SMPS

The 12 VDC 25 Amp (300W) switching mode power supply (SMPS) shown in exhibit 7 below, converts the alternating current (AC) mains to a stable 12 VDC supply for all components in the APTDS system. The DC power supply has a rating for providing sufficient current to the motor driver and other electronic components in the APTDS system while maintaining a consistent level of output performance over extended period of time. Both the external casing and internal power supply components are durable in design and include multiple protective features to ensure safe, reliable, and uninterrupted operation.

CHAPTER 4

DESIGN

4.1 CONVEYOR SYSTEM

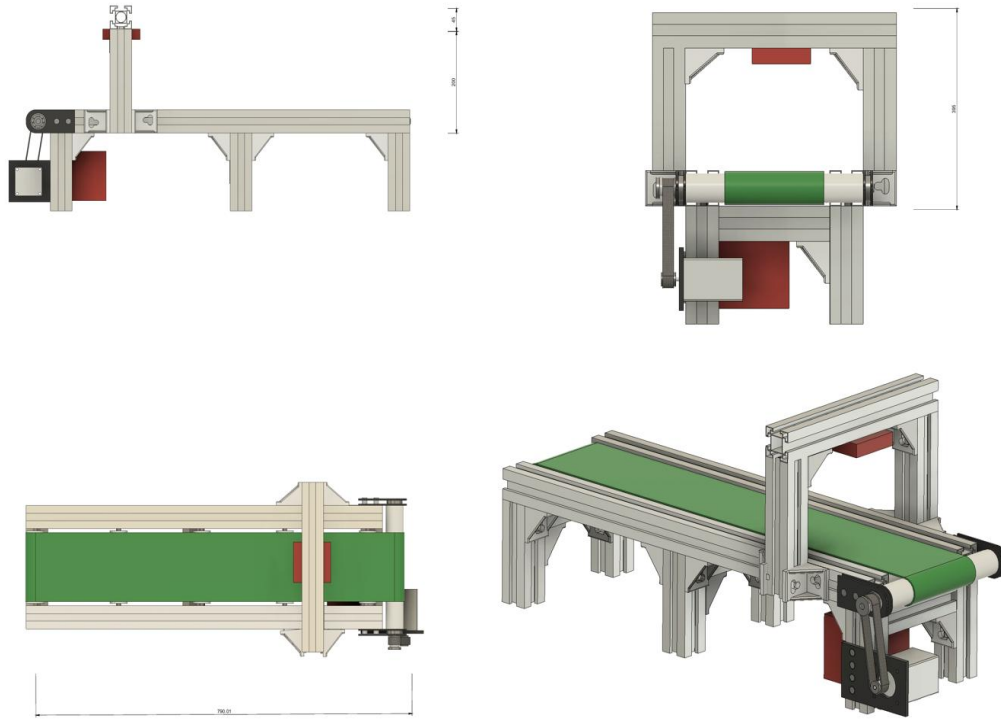


Figure 8 : Conveyor system CAD model

Figure 8: CAD Model of Conveyor System (Fusion 360)

Figure 8 shows the CAD model of the conveyor system designed using Fusion 360 for the APTDS setup. The design is presented in multiple views, including top, side, and isometric, to clearly show the structure and arrangement of components. This 700 x 390 mm conveyor is compact for prototyping applications. The structural frame on the conveyor is made from aluminum extrusions, which means it is light and easy to assemble yet strong enough to support its weight.

This conveyor also includes aluminum pulleys and rubber pulley belts- providing a reliable method of transferring power between the pulleys. The conveyor belt itself is made from polyurethane providing both traction and durability while metal ball bearings are used to reduce friction and allow the bearings to rotate easily.

A structural 3D printed DC motor mount supports the motor. The motor is mounted to the conveyor to be properly aligned with the drive wheel of the pulley, and above the conveyor is a camera to be able to see the part from an overhead view.

In general, this design incorporates all of the required elements to operate smoothly, move parts accurately and reliably through the system.

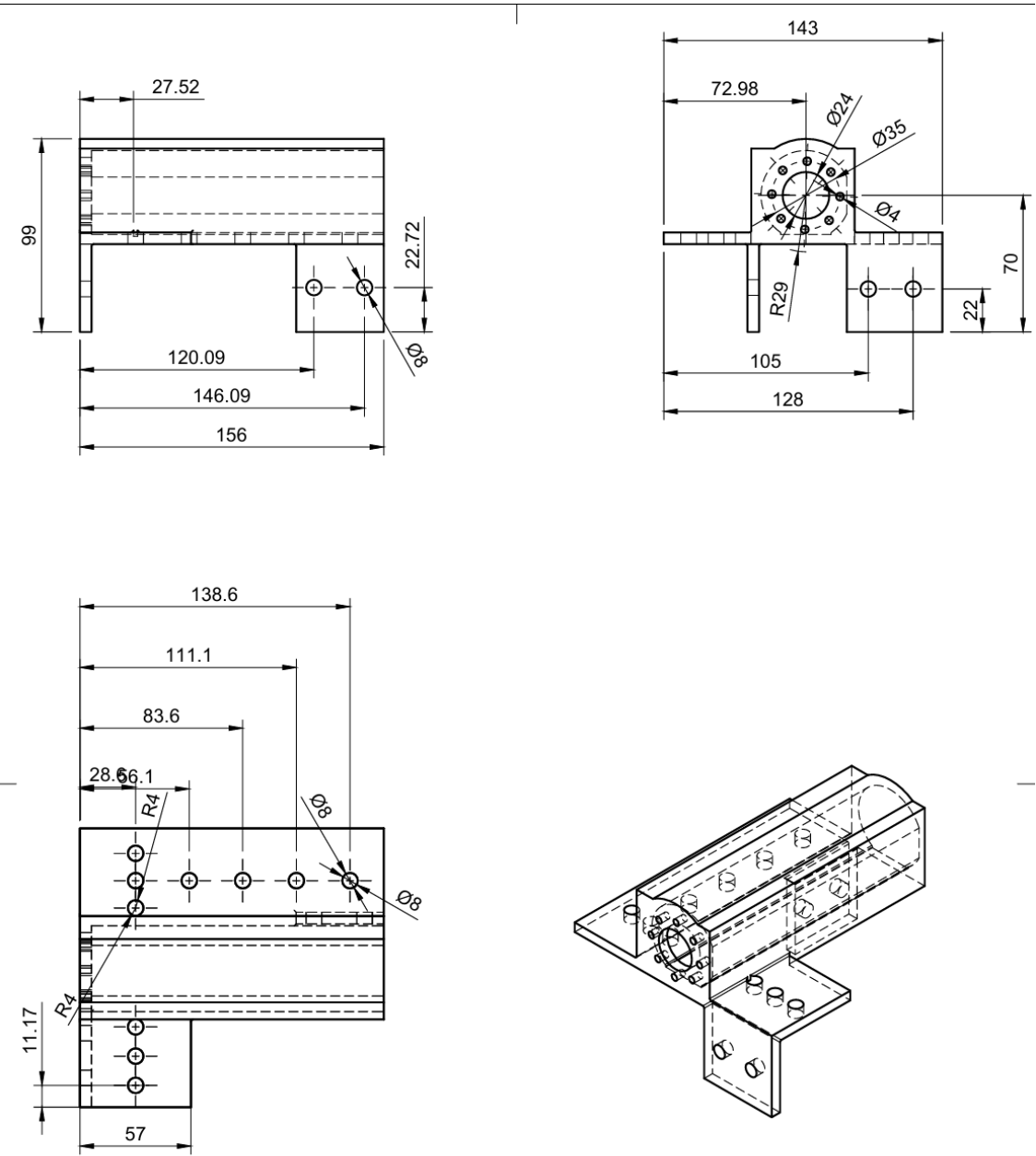


Figure 9: CAD design Motor Mount

Figure 9 shows CAD drawings of the motor mount. The motor mount is securely fastened to the DC planetary gear motor and aligned properly with the conveyor pulley by having the holes accurately drilled for mounting the motor and fixing it to the aluminium frame. This will help to provide a secure and vibration-free motor installation for the conveyor. The motor mount will be constructed from ABS plastic that can be produced using 3D printing technology, which has high strength, durability, and can easily be customized. The design will provide a stable motor support to the conveyor and allow the conveyor to operate smoothly.

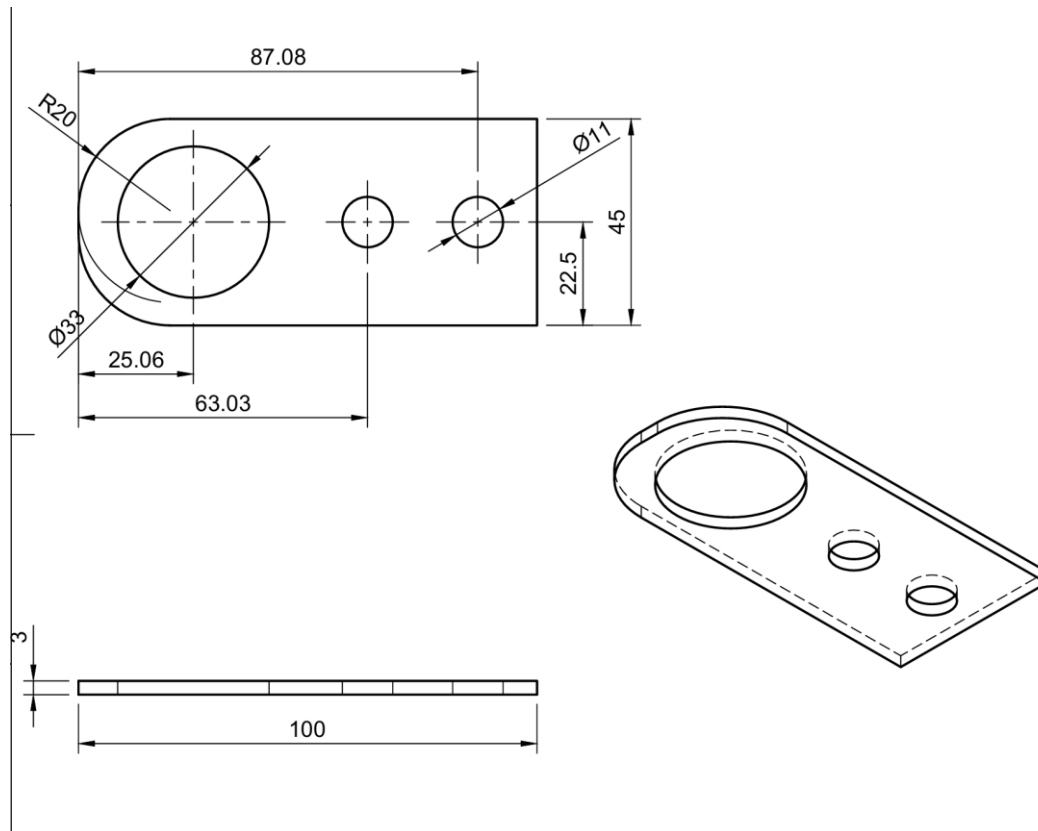


Figure 10: CAD design Bearing holder

Figure 10 provides a view of the bearing holder plate used in the APTDS conveyor system that contains a 32mm bearing. It keeps the shaft aligned so that it rotates smoothly. The bearing holder plate has two mounting holes, which allow mounting the holder plate to the conveyor frame using T-slot nuts and bolts, making it easy to install the holder plate to the frame and adjust it in position if necessary. The bearing holder plate is constructed from acrylic sheets, which keep the part lightweight but still provide adequate strength at the prototype level of development. The bearing holder

plate has a simple and compact design to provide stable support for the bearing and a reliable method of operating the conveyor system.

4.2 MATERIAL USED



Figure 11: Aluminum Profile

The aluminum profile shown in Figure 11 is a 45mm×45mm size profile that will act as the main structural frame for the conveyor because it has high strength, is light weight, and has a modular design. For this project there will be approximately 6 to 8 meters of aluminum profile used to build the base frame, support structures, and mounting structures for the conveyor and the motor. Although the aluminum profile is a light weight materials it has excellent rigidity and load-bearing ability, therefore when all components are installed there will not be any deformation of the aluminum profile due to the weight of the conveyor, motor, and other components. Additionally, there is internal structural support that adds to the overall strength of the aluminum profile therefore adding to the overall stability of the conveyor system.



Figure 12: Mild Steel Shaft

This is a photograph of a 12mm diameter, mild steel shaft used in the APTDS conveyor system with approximately 240mm length. The shaft fits perfectly inside a

32mm bearing with a 12mm inner diameter so that there is proper alignment and smooth rotation of the shaft and bearing. The shaft material (mild steel) is strong enough, durable enough, and bent/loaded well enough to support the conveyor roller and help transmit motion to the conveyor roller. The shaft provides stable rotation, proper load carrying capabilities with the conveyor system, and reliable operation of the conveyor.



Figure 12: ball bearing

The 32mm ball bearing (shown in Figure 13) is used in the APTDS conveyor system to achieve smooth, efficient rotation about the shaft. The bearing has an inner diameter of 12mm which fits precisely over the mild steel shaft and allows for perfect alignment (ie: no alignment issues) between the inner and outer diameters of the bearing.



Figure 14: PVC Roller Pipe

The PVC pipes (1½ in. diameter) that make up the rollers in the conveyor system illustrated in figure 14 are mounted on the shaft and rotate with it to allow for a smooth movement of the conveyor belt. PVC was selected for the rollers because it is lightweight, corrosion-resistant, and easy to fabricate, making it a good choice for the prototype application. While the roller is lightweight, it has sufficient strength to carry small industrial parts. Overall, the PVC rollers facilitate smooth movement of the belts,

reduce motor load, and provide an efficient means of operation for the conveyor system.

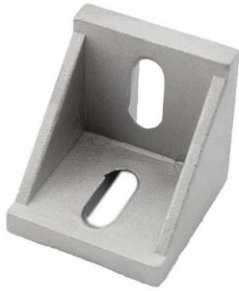


Figure 15: L-Shaped Clamps

As shown in figure 15, the L-shaped clamps used in the APTDS conveyor system provide an efficient means of connecting and supporting the aluminium profile frames. They are used at the joints of two profiles that meet at right angles to each other to ensure that they are properly aligned and remain structurally stable. The slots in the clamps make fastening them together using bolts and T-slot nuts simple and easy, resulting in a quick assembly and adjustment process. The metal clamps are strong and rigid, resulting in a strong, stable, and properly aligned conveyor structure



Figure 16: Aluminium Pulley

In the APTDS conveyor system, the aluminium pulley transmits motion from the motor to the conveyor belt through a timing or rubber belt. It is installed on a shaft and secured using set screws, which helps maintain alignment and prevents slippage. The use of aluminium allows for a lightweight pulley with strong properties, as well as providing durability and efficient power transmission. The grooved shape aids in producing and maintaining proper grip with the timing or rubber belt, as well as producing and maintaining smooth and uniform rotation. The pulley is one of the most critical components of the system, providing reliable transfer of motion and proper working components within the conveyor.



Figure 17: Timing Belt

The timing belt figure 17 used in the APTDS conveyor system transfers motion from the motor pulley to the driven pulley. The timing belt is typically made from rubber with an internal reinforcing matrix to achieve an elastomeric or flexible material; this type of material has the ability to bend and is strong. The toothed design of the timing belt allows for proper engagement with the timing pulleys, which prevents slippage and allows for accurate transfer of motion.

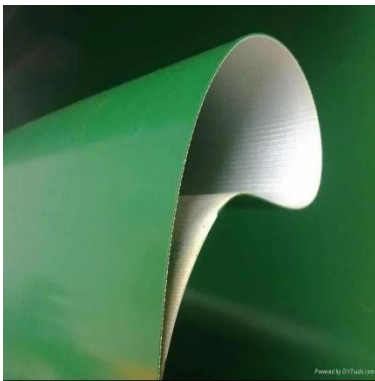


Figure 18: Polyurethane Conveyor Belt

The polyurethane conveyor belt (shown in Figure 18) is used in the APTDS conveyor and has dimensions of 1700mm long by 200mm wide. This provides a stable means of transporting components across the conveyor surface. The polyurethane material provides excellent durability, flexural resistance, and grip between the conveyor and the parts being transported, i.e.: prevents slippage. The polyurethane conveyor belt also has excellent wear resistance properties and can be used continuously. Consequently, the polyurethane conveyor belt enables smooth, reliable movement of parts across the conveyor.

4.3 WORKING MECHAINSIM



Figure 19: Driven Pulley

The APTDS system's polyurethane conveyor belt measures 1700 mm long and 200 mm wide, offering smooth and even transport of parts. Polyurethane will provide an adequate amount of strength, flexibility and traction to avoid parts from slipping off of the belt while being operated. Due to its durable qualities, the belt can withstand wear and tear from the operation and will provide reliable transport of parts for the duration of their used

The length of the timing belt can be calculated using the standard belt length equation:

$$L = 2C + \frac{x(D+d)}{2} + \frac{(D+d)^2}{4C} \dots\dots\dots(1)$$

Where:

L = Belt length

C = Center distance (100 mm)

D = Diameter of larger pulley (44 mm)

d = Diameter of smaller pulley (44 mm)

Since both pulleys are of equal size ($D = d$), the equation simplifies to:

$$L = 2C + \pi D \dots\dots\dots(2)$$

Substituting the values:

$$L = 2(100) + \pi(44) \dots\dots\dots(3)$$

$$L = 200 + 138.23 \dots\dots\dots(4)$$

$$L \approx 338.23 \text{ mm} \dots\dots\dots(3)$$

The actual length of the required belt is 338 mm. The 6 mm belt width will provide adequate strength/stability (power transmission) but will have no effect on the calculation of the length. Both pulley's diameters are the same, so the other pulley will rotate at the same speed as the first (1:1 ratio), thereby creating a uniform rotation/smooth operation for the entire conveyor system.

Conclusion

The design of the APTDS system has been developed with an emphasis on the use of simple designs, sturdy strength, and application approaches. The construction materials used for the conveyor system have all been sourced from a single supplier, aluminum extrusions. The base of the system uses aluminum extrusions to provide a structural framework that is both strong (much like steel) and light in weight. With the design of components such as the motor mount, bearing holder, and rollers, special attention has been given to the alignment of components, the operation of motors to produce smooth and stable operational characteristics.

Using mild steel, PVC, and polyurethane, the design optimizes the balance between durability/cost/performance. As a result, the APTDS system is appropriate for use in prototype-level industrial environments. The operating principle of the conveyor, particularly the construction of the timing belt driven pulley mechanism, ensures that the transfer of motion from the drive motor to the conveyor is accomplished without slippage and with precision. Due to the appropriate design and selection of components such as pulley size, shaft length, bearing diameter, and type of belt, together with the use of pulleys of equal size, the system allows for the synchronization of component movement.

Overall, the APTDS system has a design that is functional, reliable, and requires little maintenance, establishing a solid foundation upon which to build the automated and detection capabilities of the APTDS system

CHAPTER 5

PROGRAMMING

5.1 SYSTEM PROGRAM ARCHITECTURE

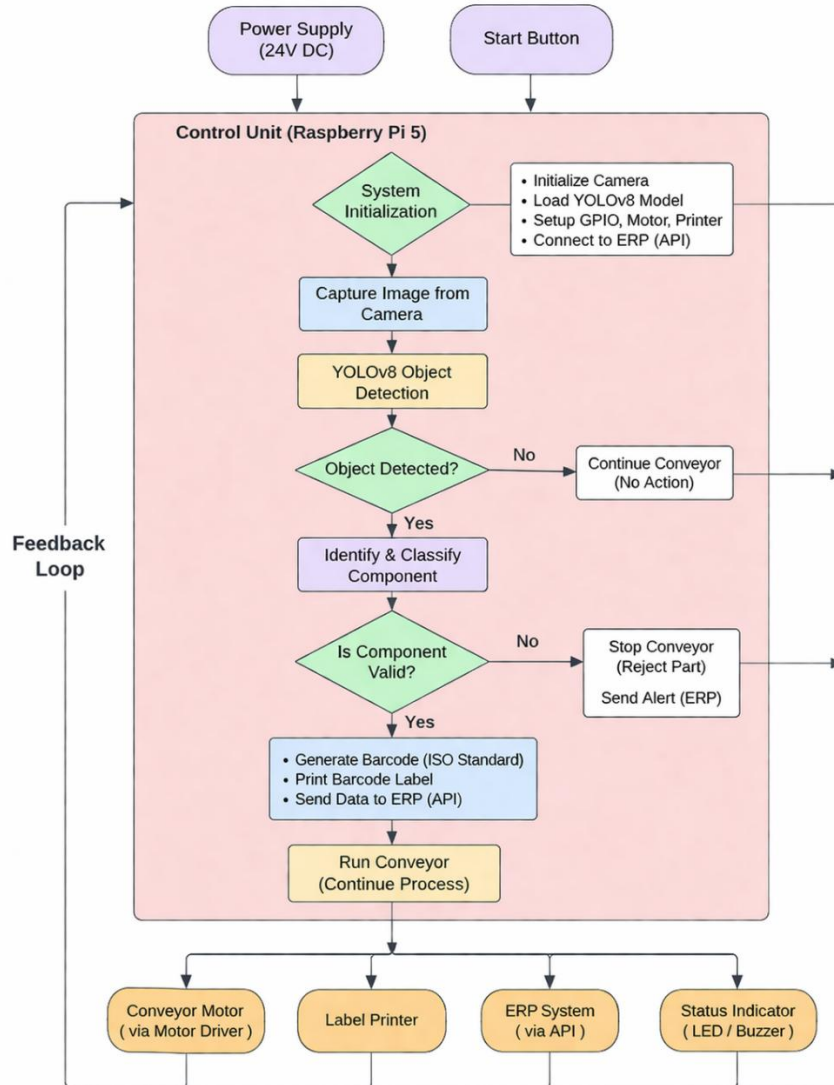


Figure 20: Program Flow Diagram of APTDS System

The diagram below (Figure 20) shows how the automated part traceability and data synchronization (APTDS) operates. It demonstrates the processing of input, detection, and controlling of output in a continuous cycle (or loop) by this system. This all begins with the initializing of the APTDS. The Raspberry Pi 5 runs through all required initializations including to turn on the camera, load the YOLOv8 algorithm, configure

the GPIO pins and establish communication with the enterprise resource planning (ERP) system.

The APTDS will check for the presence of any component following the completion of the detection process. No object detected will cause the conveyor to continue running in a forward motion. If an object is detected it will be identified and classified; there will then be an additional step to validate that the identified and classified part is within the APTDS defined specifications. If the identified and classified part does not meet the APTDS defined specifications the conveyor will stop, the part will be rejected, and an alert will be sent to the ERP System. If the identified and classified part does meet the APTDS defined specifications, the APTDS will generate and print the label with the ISO standard barcode and at the same time will update the ERP database through the API.

Once all of the required label and data synchronization procedures have been performed and completed, the conveyor will continue to operate allowing movement of the part through the system. An automated feedback loop is used for continual monitoring and instantaneous decision-making. A structured flow provides for the accurate identification and verification of activities, prevents the occurrence of errors, and allows for seamless integration between the physical and digital processes, making the APTDS efficient, dependable, and fully automated.

5.2 AI Model Development Using YOLOv8

The AI model used in the APTDS system was developed using the YOLOv8 framework, which is a well-known and fast and accurate real-time object detection framework.

After the images have been collected, the next step is to label the data absolutely annotated with bounding boxes and corresponding class labels; in this process each object is marked with a bounding box in the image and a specific class label is assigned. The annotations are the basis of the model's learning process, as each nut and bolt will be assigned to a specific class. This is extremely important for the model to differentiate between the various types of objects in the images. This is why it is essential for everyone involved in the data annotation process to be diligent in properly

labeling and accurately labeling every nut and bolt; if they are not properly labeled, the model is likely to classify the nuts and bolts as being the same. There will be no misclassifications or poor learning of object differences if the data is properly and consistently annotated; therefore, extra care is taken to ensure accurate data annotation.

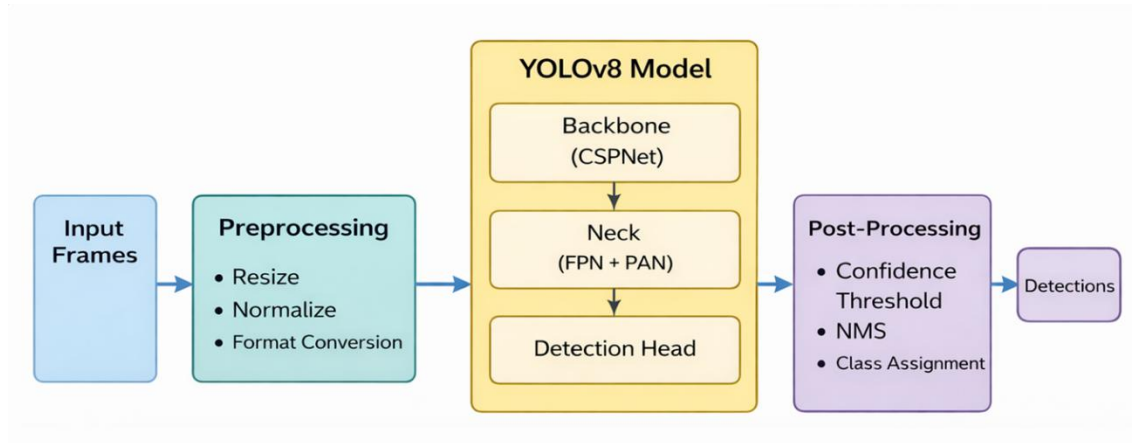


Figure 21: YOLOv8 Architecture

The development of the AI model starts with data collection by capturing a large number of images of industrial components (M5, M6, M8 and M10 nuts and bolts) from the cameras on the developing APTDS system under different lighting conditions, angles, distances and with different backgrounds to provide a dataset that is representative of the real working environment. The diversity of images helps the model to learn and improve the ability to detect the component, even if the position, orientation or lighting conditions are different in actual operation.

After the images have been collected, the next step is to label the data absolutely annotated with bounding boxes and corresponding class labels; in this process each object is marked with a bounding box in the image and a specific class label is assigned. The annotations are the basis of the model's learning process, as each nut and bolt will be assigned to a specific class. This is extremely important for the model to differentiate between the various types of objects in the images. There will be no misclassifications or poor learning of object differences if the data is properly and consistently annotated; therefore, extra care is taken to ensure accurate data annotation.

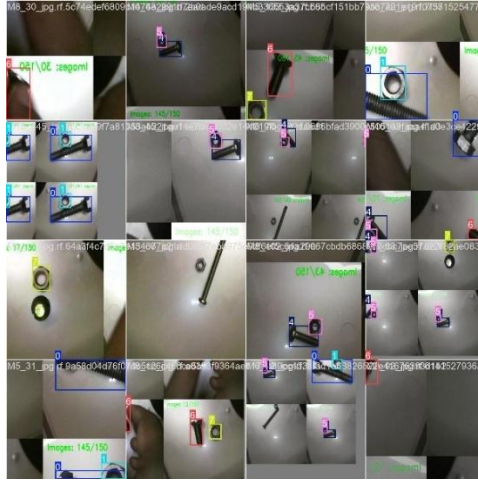


Figure 22: Trained Model

In the last step, the model is validated using new, never-before-seen images to determine whether it performs as desired in both present conditions and future situations. The images shown in Figure 22 successfully demonstrate that the model accurately detects and classifies all nuts and bolts by outputting bounding boxes and confidence scores, which indicates that it can properly classify and detect components in real time.



Figure 23: Validation Dataset

Next, the labelled dataset is split into a training dataset and a validation dataset figure 23 to provide additional performance information on how well the model will function using a dataset that does not contain any images previously used during training. The validation data set contains numerous types of images showing nuts and bolts at

different angles, positions, and arrangements with their corresponding labels. These images are used to determine how well the model will detect and identify parts in real-world conditions.

5.2.1 PROGRAMMING CODE

Purpose:

This code describes how the major features of the APTDS system are delivered through the integration of real-time object recognition, ERP data management, and bar code generation technologies. Through the delivery of automated component identification, reliable data capturing, and seamless integration of the physical system with the digital dashboard, the primary goal of this document is to provide a description of how the software that makes up the backend of the APTDS system is implemented.

```
from flask import Flask, render_template, Response, request, redirect, session, jsonify
```

```
import sqlite3
```

```
import barcode
```

```
from barcode.writer import ImageWriter
```

```
from datetime import datetime
```

```
import os
```

```
import cv2
```

```
from ultralytics import YOLO
```

```
import threading
```

```
import time
```

```
app = Flask(__name__)
```

```
app.secret_key = "erp_secret"
```

```

DB = "database.db"

ENGINE_MAP = {

    "M10": "Common rail engine",

    "M8": "Mechanical rail engine",

    "M6": "Mechanical rail engine"

}

def init_db():

    conn = sqlite3.connect(DB)

    c = conn.cursor()

    c.execute("""

CREATE TABLE IF NOT EXISTS parts (

    id INTEGER PRIMARY KEY AUTOINCREMENT,

    part_name TEXT,

    engine_type TEXT,

    batch_number TEXT,

    barcode_path TEXT,

    created_date TEXT

)

""")

    conn.commit()

    conn.close()

```

```

def insert_part(label):

    engine_type = ENGINE_MAP.get(label, "Unknown")

    batch_number = "APT" + datetime.now().strftime("%Y%m%d%H%M%S")

    barcode_class = barcode.get('code128', batch_number, writer=ImageWriter())

    barcode_path = f"static/barcode/{batch_number}"

    barcode_class.save(barcode_path)

    conn = sqlite3.connect(DB)

    c = conn.cursor()

    c.execute("""

        INSERT INTO parts

        (part_name, engine_type, batch_number, barcode_path, created_date)

        VALUES (?, ?, ?, ?, ?)

    """, (

        label,

        engine_type,

        batch_number,

        barcode_path + ".png",

        datetime.now().strftime("%Y-%m-%d %H:%M:%S")

    ))

    conn.close()

    print("Inserted:", label, engine_type)

model = YOLO("/home/raspi5/pp2/best.pt")

```

```

camera = cv2.VideoCapture(0)

camera.set(cv2.CAP_PROP_FRAME_WIDTH, 640)

camera.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)

latest_frame = None

detected_objects = set()

no_object_counter = 0

RESET_AFTER = 12

def detection_thread():

    global latest_frame, detected_objects, no_object_counter

    while True:

        ret, frame = camera.read()

        if not ret:

            continue

        latest_frame = frame.copy()

        results = model.predict(frame, conf=0.5, verbose=False)

        current_labels = set()

        for r in results:

            if r.bboxes is not None:

                for box in r.bboxes:

                    cls_id = int(box.cls[0])

                    label = model.names[cls_id]

```

```

        current_labels.add(label)

        if label not in detected_objects:

            insert_part(label)

            detected_objects.add(label)

    if len(current_labels) == 0:

        no_object_counter += 1

    else:

        no_object_counter = 0

    if no_object_counter > RESET_AFTER:

        detected_objects.clear()

    time.sleep(0.2)

def generate_frames():

    global latest_frame

    while True:

        if latest_frame is None:

            continue

        frame = latest_frame.copy()

        results = model.predict(frame, conf=0.5, verbose=False)

        for r in results:

            if r.bboxes is not None:

                for box in r.bboxes:

                    cls_id = int(box.cls[0])

                    label = model.names[cls_id]

```

```

        x1, y1, x2, y2 = map(int, box.xyxy[0])

        cv2.rectangle(frame, (x1, y1), (x2, y2), (0,255,0), 2)

        cv2.putText(frame, label, (x1, y1-10),

                    cv2.FONT_HERSHEY_SIMPLEX,

                    0.7, (0,255,0), 2)

    _, buffer = cv2.imencode('.jpg', frame)

    frame = buffer.tobytes()

    yield (b'--frame\r\n'

          b'Content-Type: image/jpeg\r\n\r\n' + frame + b'\r\n')

@app.route('/')
def index():

    return render_template(

        "index.html",

        today=datetime.now().strftime("%d-%m-%Y"),

        login_time=session.get("login_time"),

        user=session.get("user")

    )

@app.route('/video')
def video():

    return Response(generate_frames(),

                    mimetype='multipart/x-mixed-replace; boundary=frame')

@app.route("/api/parts")
def get_parts():

    conn = sqlite3.connect(DB)

    c = conn.cursor()

    c.execute("SELECT FROM parts ORDER BY id DESC LIMIT 20")

```

```

rows = c.fetchall()

conn.close()

data = []

for r in rows:

    data.append({

        "id": r[0],

        "size": r[1],

        "engine": r[2],

        "batch": r[3],

        "barcode": r[4],

        "date": r[5]

    })

return jsonify(data)

@app.route("/login", methods=["POST"])

def login():

    session["user"] = request.form["name"]

    session["login_time"] = datetime.now().strftime("%H:%M")

    return redirect("/")

@app.route("/logout")

def logout():

    session.clear()

    return redirect("/")

if __name__ == "__main__":

    os.makedirs("static/barcode", exist_ok=True)

    init_db()

    t = threading.Thread(target=detection_thread)

```

```
t.daemon = True

t.start()

app.run(host="0.0.0.0", port=5000, threaded=True)
```

Description:

This document describes how the backend of the APTDS system is built and configured using Flask to create a complete solution consisting of multiple modules such as camera input, YOLOv8 object detection, database storage, and bar code generation into a unified workflow. After this information is used to create and save the associated bar code, it is also entered into an SQLite database for later retrieval.

The APTDS uses a separate background detection thread to process frames continuously without interrupting the main application. The web interface displays live video of the camera view as well as real-time data from the SQLite database as obtained through API calls to the database. Lastly, all logic to prevent duplicative entries through tracking of objects already detected is included in the code. Overall, this document provides a comprehensive overview of how the APTDS is implemented, including providing for automated component identification, real-time monitoring of detected components, and reliable handling of detected component data as they are being processed through the APTDS and helps to provide the basis of the APTDS.

5.3 ERP Integration



Figure 24: ERP system

The APTDS and ERP Setup is best viewed in Figure 24. To manage real time data, as soon as the Raspberry Pi processes the part, the camera detects the part and then sends the detected part information from the Raspberry Pi to the APTDS and ERP and saves the data as JSON via an HTML based ERP interface. This allows the data interface to remain lightweight, fast, and easy to maintain, as the system does not require complex database systems.

The ERP interface provides an easy to view and organized layout with detail of each part's name/type, detection time, and status. Each of the detected, completed, and tracked part has automatically generated a unique batch number to assist in the tracking of component parts through the system. As the time between detection and data created is instantaneous, the overall system becomes more efficient with a reduction in possible errors.

Another added benefit of Figure 24 is that the user can visually monitor the real time camera feed of the system detecting parts.

5.3.1 Barcode

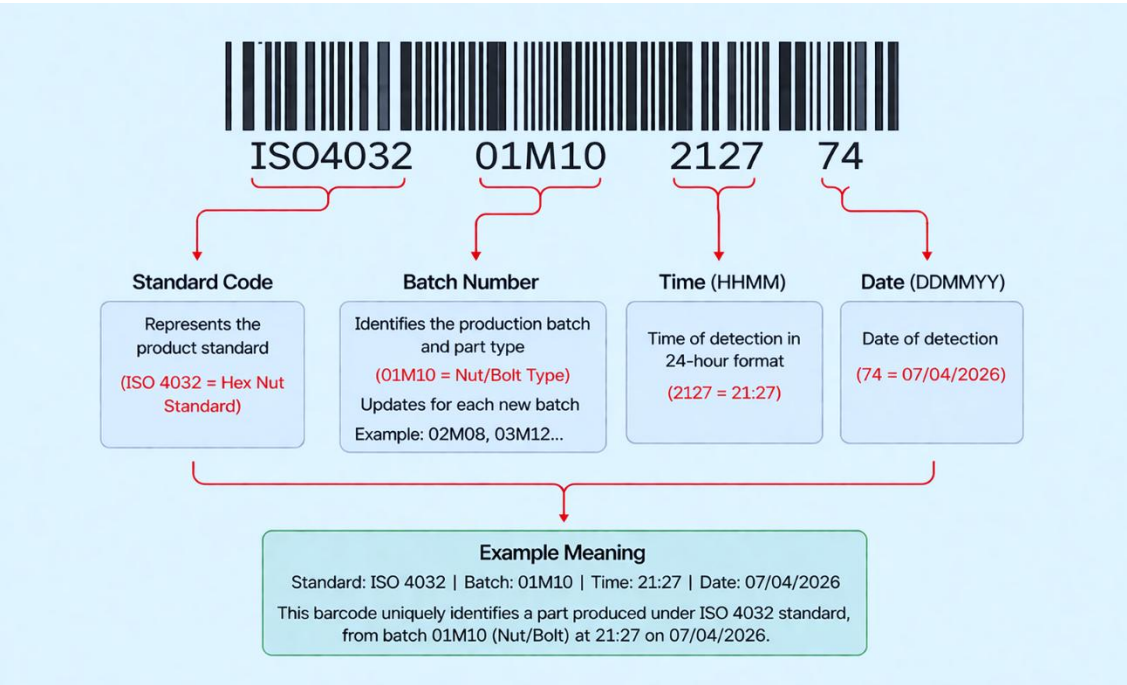


Figure 25: Barcode Format Explanation

Nonetheless, this combined piece of information will also tell an operator what size or type the component is, and by scanning its barcode, they can effectively pull up every part made in that particular batch. This will lead to higher levels of confidence in the part being pulled, resulting in fewer mistakes on the job site or during assembly.

The last part of the barcode is represented by a series of "X" characters. The "X" is being utilised primarily as a way to represent that the particular code will continue to grow as we produce more and more parts within our organisation (there would be one X code for every level or code at the beginning of our future numbering system, which will be added to as we produce more components). Ultimately, creating a universal and globally recognised barcode for all of the APTDS's parts will allow us to continue expanding as an organisation without compromising our reputation or the integrity of our quality assurance program.

The last part of a barcode, 2127, indicates the exact time a part was detected and processed, in 24-hour terms (21:27). This allows tracking of the order of operations and when the part passed through the system. The last part of the barcode, 74, indicates the date (07/04/2026) that records the order of manufacture for the part, even though it has been written in short format.

Together, all of these components create a complete identity for each component of a system. This allows for identification of what the component is, the batch of components it is a part of, and the exact time that it was processed. This structure helps to provide a more efficient, reliable, and less error prone system, as well as improves the ability to trace components back to their origins. In addition, this design of barcodes will connect physical components to digital records, making the APTDS system as efficient and manageable as possible.

5.3.2 HTML CODE

Purpose:

Just like the APTDS system ERP Dashboard, this HTML file is aimed at an easy-to-use interface that displays realtime detection data, a live camera feed, and stored part information for the ERP system. The purpose of this dashboard is to provide an

operator with an easy-to-use interface to monitor the activities on a continuous and immediate basis.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>APTDS ERP</title>
```

```
<style>
```

```
body{font-family:Arial;margin:0;background:f4f4f4;}
```

```
.topmenu{
```

```
background:4c74b9;
```

```
padding:10px;
```

```
color:white;
```

```
display:flex;
```

```
gap:30px;
```

```
font-weight:bold;
```

```
}
```

```
.container{padding:20px;}
```

```
.flexbox{display:flex;justify-content:space-between;}
```

```
.left-panel{width:45%;}
```

```
.right-panel{
```

```
width:45%;
```

```
border:2px solid gray;
```

```
height:250px;
```

```
background:white;
```

```
}
```

```
.orange-box {  
background:ea7b2c;  
color:white;  
padding:10px;  
margin-bottom:10px;  
}  
  
.yellow-box {  
background:ffc000;  
padding:10px;  
margin-bottom:10px;  
}  
  
table {  
width:100%;  
border-collapse:collapse;  
margin-top:20px;  
}  
  
th {  
background:6ea94d;  
color:white;  
padding:10px;  
}  
  
td {  
padding:8px;  
border:1px solid ddd;  
text-align:center;
```

```

}

</style>

</head>

<body>

<div class="topmenu">

<div>Home</div>

<div>File</div>

<div>Tools</div>

<div>Edit</div>

<div>Save</div>

<div>Actions</div>

<div>Help</div>

</div>

<div class="container">

<div class="flexbox">

<div class="left-panel">

{% if user %}

<div class="orange-box">

Log In time {{ login_time }}

<br>

<a href="/logout" style="color:white;">Log out</a>

</div>

{% else %}

<form action="/login" method="POST">

<input name="name" placeholder="Incharge Name" required>

```

```

<button type="submit">Login</button>

</form>

{% endif %}

<div>

<b>Date</b> {{ today }}

</div>

<div class="yellow-box">

<b>Incharge name</b> {{ user }}

</div>

</div>

<div class="right-panel">



</div>

</div>

<table>

<tr>

<th>S.No</th>

<th>Part name</th>

<th>Batch Number</th>

<th>Barcode</th>

<th>Date</th>

</tr>

<tbody id="partsTable"></tbody>

</table>

</div>

```

```

<script>

function loadParts(){

fetch("/api/parts")

.then(res=>res.json())

.then(data=>{

let table=document.getElementById("partsTable");

table.innerHTML="";

data.forEach((p,index)=>{

table.innerHTML+=`

<tr>

<td>${index+1}</td>

<td>${p.name}</td>

<td>${p.batch}</td>

<td></td>

<td>${p.date}</td>

</tr>

setInterval(loadParts,2000);

loadParts();

</script>

</body>change in this also

</html> do it for thsi also

```

Description:

The front-end portion of the dashboard for the APTDS system is written using HTML, CSS, and JavaScript in the actual source code. The dashboard consists of multiple sections including a top menu, user info panel, live camera feed, and a data table. In general, the left section displays the login information for the logged-in user, today's

date and the name of the person in charge and the right section displays the live video stream from a camera using Flask Routing.

The data table in the lower part of the ERP Dashboard updates dynamically to provide the user with current information about all of the detected parts that have been discovered by the APTDS System (Part Name, Batch Number, Barcode Image and date). The data from the backend API (/api/parts) is fetched via JavaScript and the data is returned to update the parts in the data table. The data table updates every two seconds so the user will not need to refresh the page to see updated data. Overall, the dashboard has an organized, clean and easy to read user interface using a simple color palette, organized layout and logical flow to connect the user with the backend of the APTDS system. The dashboard will provide a real-time monitoring capability and a way to interface smoothly with the APTDS system.

Conclusion

The APTDS system demonstrates how several different technologies can be combined for both ease of automation as well as increased intelligence in manufacturing businesses through the use of YOLOv8 for detection and Raspberry Pi for process control. This allows for real-time identification of parts and the ability to perform all actions automatically without the need for human intervention. In turn, this reduces human error, saves time, and increases the efficiency and reliability of the manufacturing process.

Additionally, the ERP-integrated barcode systems allow for streamlined tracking of each component. Each component is logged with complete tracking information so that any one particular component can be identified, when it was completed, and where it should be stored/used. The live visualisation dashboard also provides the user with an extremely clear view of all aspects of the system. Overall, the APTDS system is very basic, efficient, and has the ability to evolve to provide even greater benefits to industries as their continued growth occurs during the next few years.

CHAPTER 6

RESULTS AND ANALYSIS

6.1 AI MODEL ANALYSIS

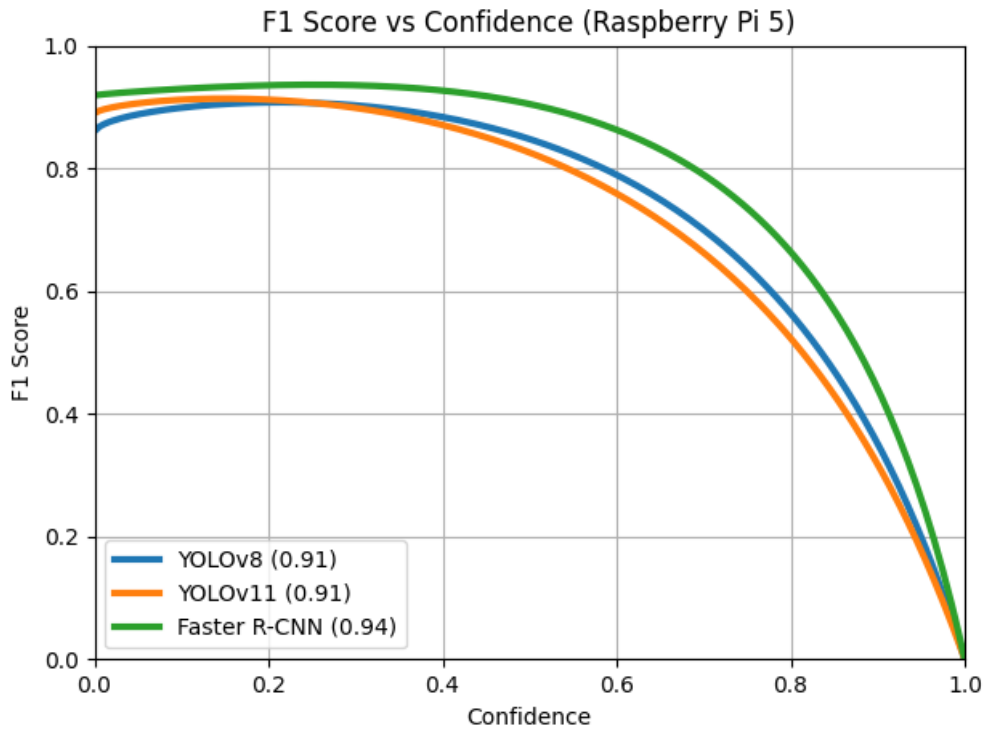


Figure 25: F1 Score vs Confidence

The graph shows how the F1 score of the models tested at various confidence levels vary from one another. The F1 score is useful because it allows us to look at both precision and recall in a balanced view. Looking at the F1 score graph, we can see that all three models showed good performance at roughly the same confidence levels, particularly within the interval from 0.2 to 0.6, where their F1 scores remain constant and high. Faster R-CNN scored just slightly higher than the other models with a score of near 0.94, confirming this model has a better balanced performance regarding its ability to identify and also avoid false detections.

As the confidence level continues to rise above some threshold value, therefore the F1 scores for the models begin to decrease. Therefore, higher confidence levels create a

more strict model resulting in the model missing a valid object. In layman's terms, the model becomes too careful to identify the object and therefore misses objects that it would have identified prior to creating the higher confidence level threshold. Therefore, this observation demonstrates the importance of determining an optimal value for your confidence level rather than simply using a higher value for your confidence level.

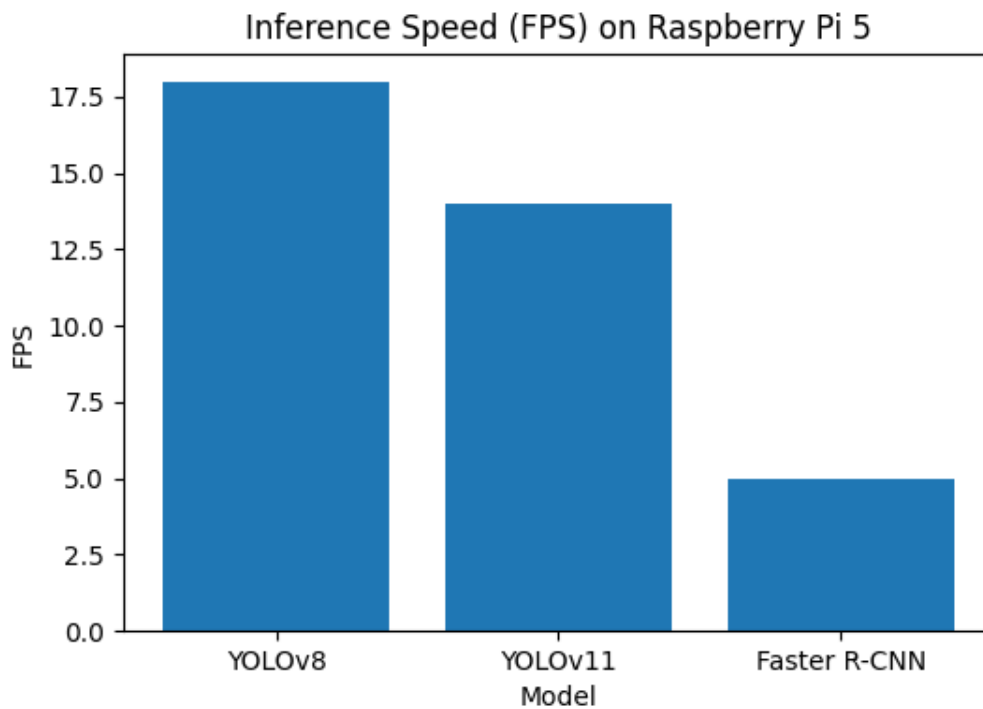


Figure 26: Inference Speed (FPS) on Raspberry Pi 5

The Raspberry Pi 5 performance comparison of all three models is very important since the Raspberry Pi system runs in real-time on the conveyor. YOLOv8 runs the fastest at about 18 FPS, meaning that YOLOv8 will have little to no time delay in processing the frames quickly and continuously. YOLOv11 is next at about 14 FPS while the speed of Faster R-CNN running will be approximately 5 FPS.

From a practical perspective, YOLOv8 can maintain smooth, real-time detection of moving components along the conveyor. In contrast, due to the lower speed of Faster R-CNN, it may have to deal with detection delays or missed detections from moving components. Therefore, it is clearly obtained that lightweight models will be the preferred model for embedded systems such as Raspberry Pi.

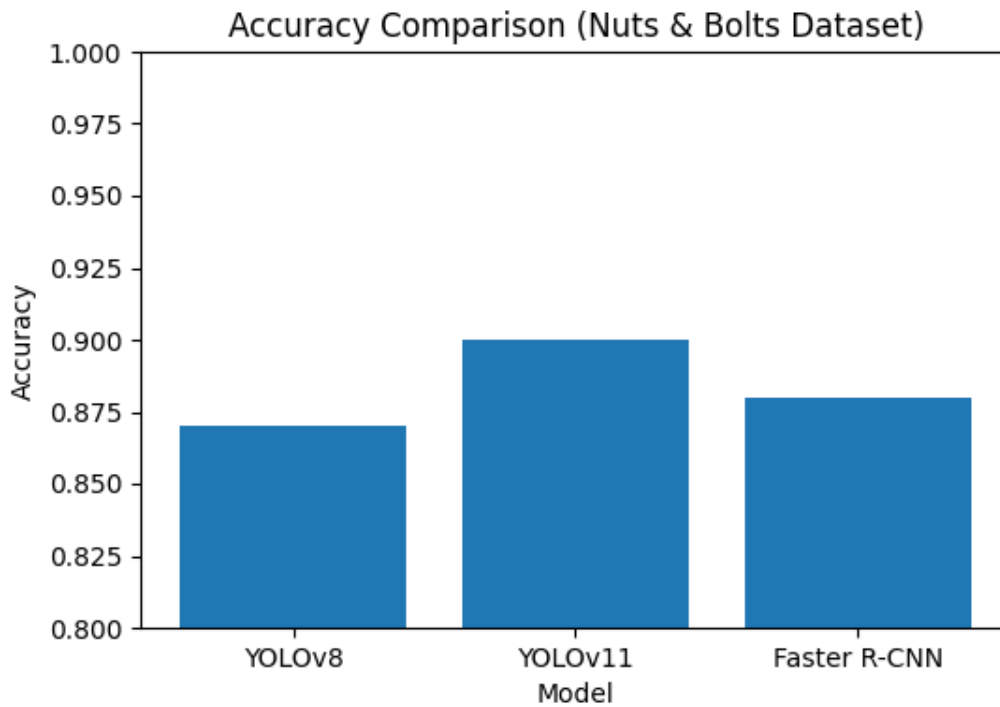


Figure 27: Accuracy Comparison

The overall accuracy of each model is illustrated in figure 27 with the highest accuracy being achieved by YOLOv11 (90%), much higher than either Faster R-CNN (88%) or YOLOv8 (87%); thus all models appear capable of accurately identifying components reliably regardless of how small a difference exists between them.

Although YOLOv11 does provide marginally better accuracy, the difference is not substantial; rather it would demonstrate that in real world scenarios, other key performance metrics other than accuracy (e.g., speed) are important when evaluating various models for practical implementations. Therefore, while Yolo v11 is more accurate than the other models evaluated; because it has about the same overall accuracy and some measure lower speed, it could be used in a similar or possibly better purpose than either model by providing an overall higher overall speed than another model; thus sacrificing some bit of max accuracy (as much as accuracy has already been achieved from accurate identification).

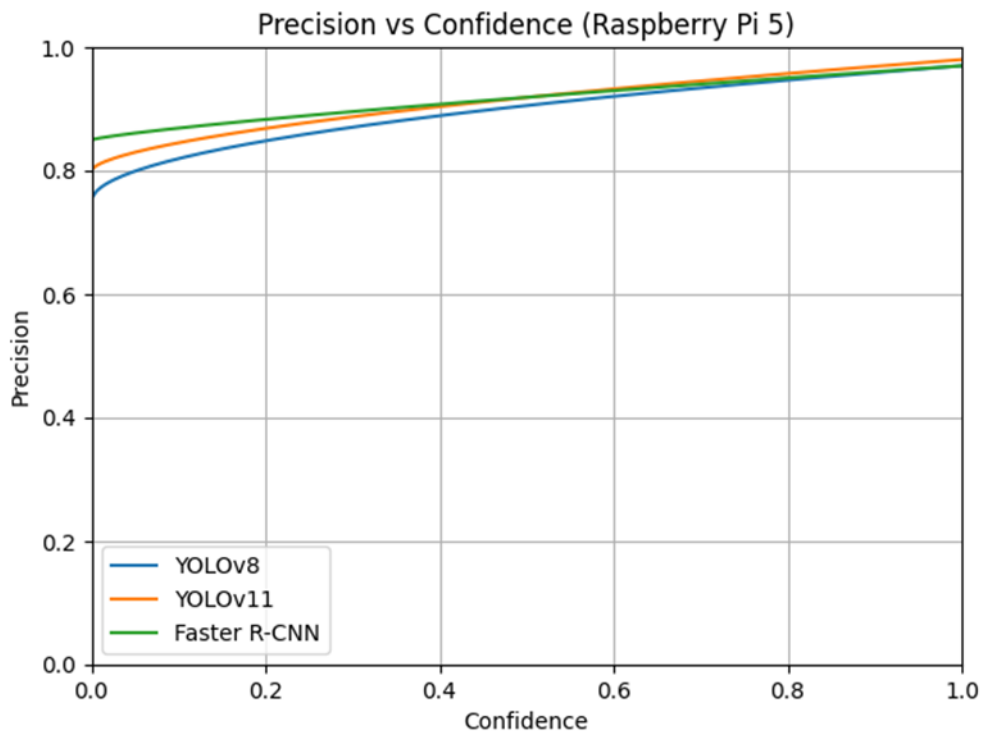


Figure 28: Precision vs Confidence

Figure 28 depicts the relationship between precision and levels of confidence. Precision indicates how many instances of an object detection can be considered correct. Increased confidence produces a higher level of precision; this is to be expected as the model becomes more selective over time by eliminating false positives from its predictions.

As shown in Figure 28, Faster R-CNN produces slightly higher values of precision among similar levels of confidence than either of the YOLO models, indicating that it is doing a good job at predicting objects. The YOLOv11 model also exhibits high levels of precision, particularly at higher levels of confidence; therefore, this demonstration indicates all three models have the ability to provide trustworthy detections, but achieving a high level of precision could result in the omission of an object from being detected due to the cooler prior experience with the detection.

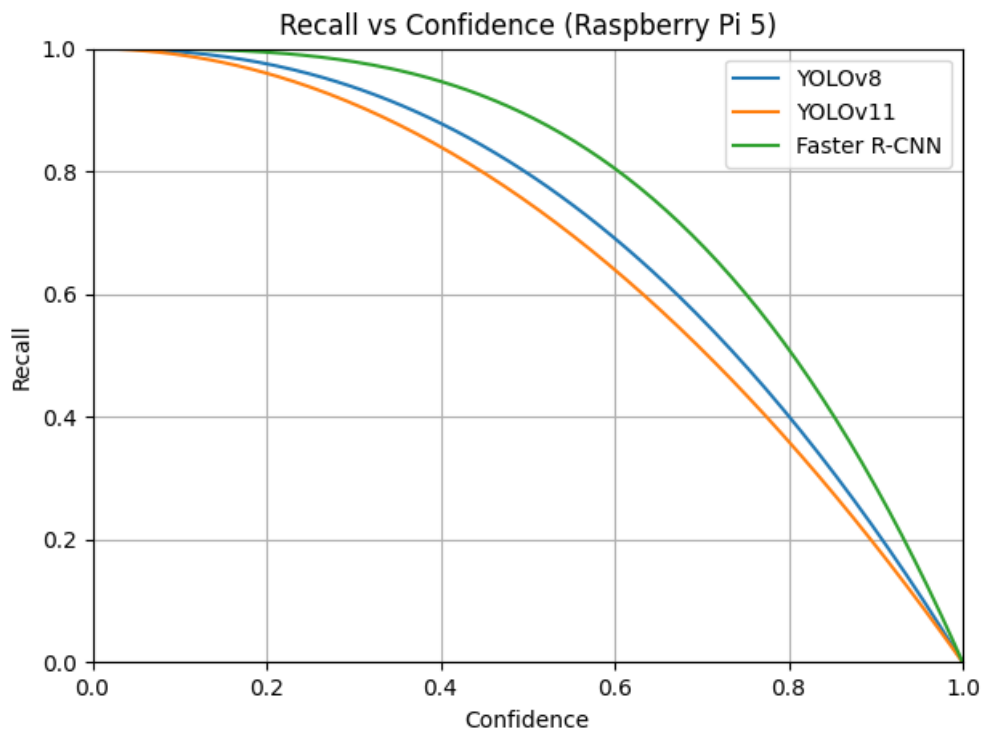


Figure 29: Recall vs Confidence

Recall refers to how many true detections of actual objects were counted. As we increase the confidence threshold (essentially our "cut-off" point for confidence, above which an object is detected) for the number of detections, the number of true detections will decrease because the model can no longer detect things that are uncertain.

Faster R-CNN has the highest level of recall compared to the faster versions of YOLO (YOLOv8/YOLOv11), meaning it will find more objects than either of those YOLOs; however, both YOLOv8 and YOLOv11 experience a more significant drop in recall at higher levels of confidence. This illustrates the tradeoffs between detection and accuracy.

Overall Analysis

All models have their respective strengths and weaknesses based on the performance metrics evaluated. The fastest overall detection quality (highest F1 score, precision, and recall) is provided by Faster R-CNN; however, the primary drawback of this model is its slow operation, which makes it unsuitable for real-time applications.

In contrast, YOLOv8 gives a great balance between speed and performance; it operates at a significantly higher speed than the others while still providing acceptable levels of detection quality and accuracy. YOLOv11 provides improved accuracy compared to YOLOv8 at a very slight decrease in operating speed.

In the case of a system like APTDS, where components move constantly on a conveyor system, speed is extremely important; therefore, YOLOv8 would be the best model because it will give the ability for fast, continual, and reliable real-time detection. The overall analysis indicates that a selection of the best model for your application will depend not only on the accuracy of the models, but also on their operating characteristics in the real-world.

6.2 YOLOV8 PERFORMANCE EVALUATION AND ANALYSIS

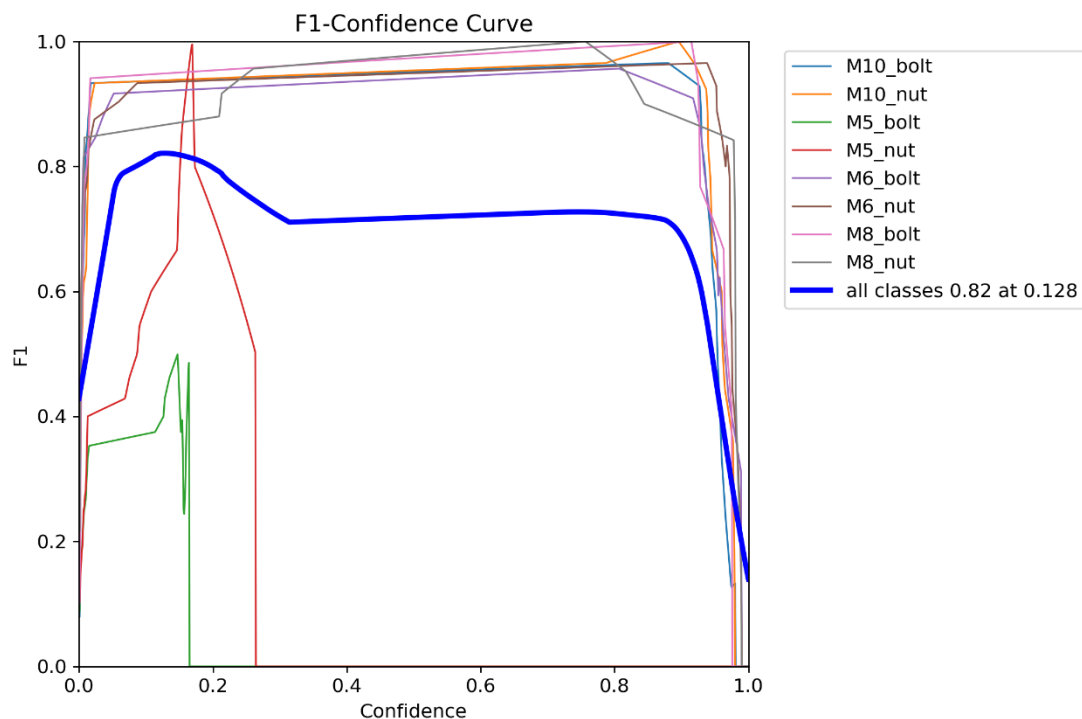


Figure 30: F1–Confidence Curve

The overall F1 Score for all classes (the solid line) indicates high model performance (best F1 score ~0.82) at ~0.13 confidence. When the model has low confidence, it detects (detects more objects) but makes numerous incorrect detections also. As confidence increases, the model becomes much more selective (accurate) with fewer

objects being detected. As confidence increases, the overall detection coverage will decrease.

The graph shows that larger classes (M10) are detected consistently whereas the smaller classes (M5) exhibit increased variability in terms of object detectability. Overall, this curve aids in determining the best confidence threshold for obtaining balanced model performance.

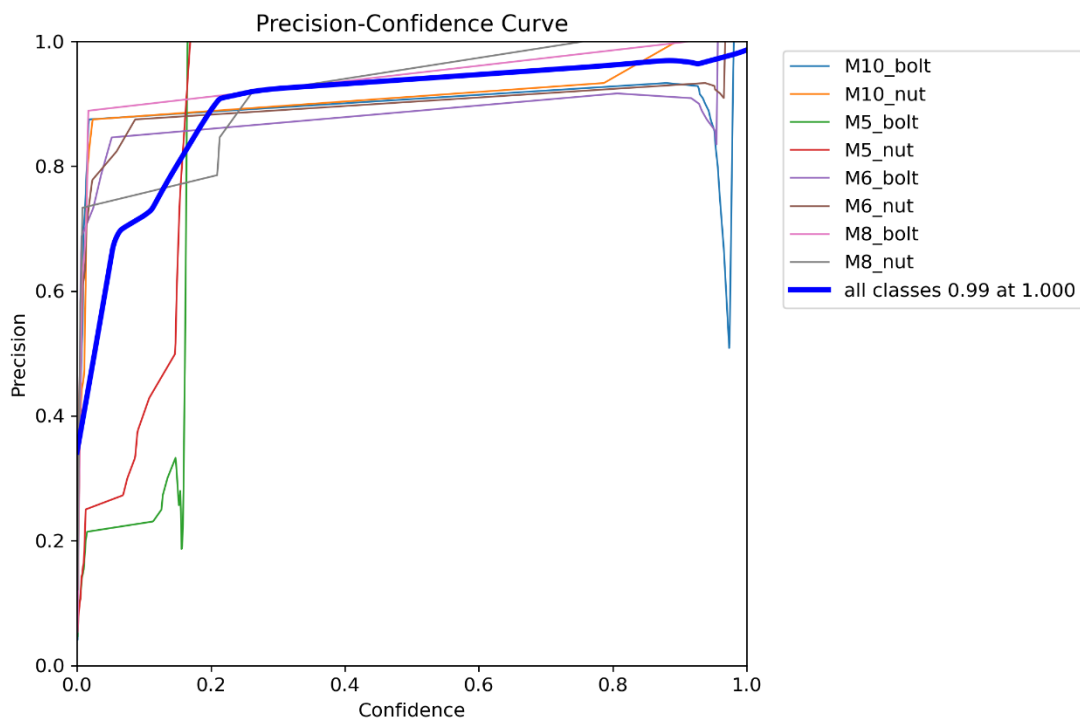


Figure 31: Precision–Confidence Curve

Graphically, Figure 31 shows quality of intent with increasing degree of quality. At a high degree of confidence, the intent quality reaches almost 0.99 which indicates that almost all detections are correct so the model can be trusted at this point. Conversely, at a lower level of confidence, the intent quality decreases slightly as the model tries to detect more objects (both with known and unknown intent) which is anticipated in any object detection system. The smooth increasing slope illustrates that the YOLOv8 model has consistently produced high quality predictions across varied levels of confidence.

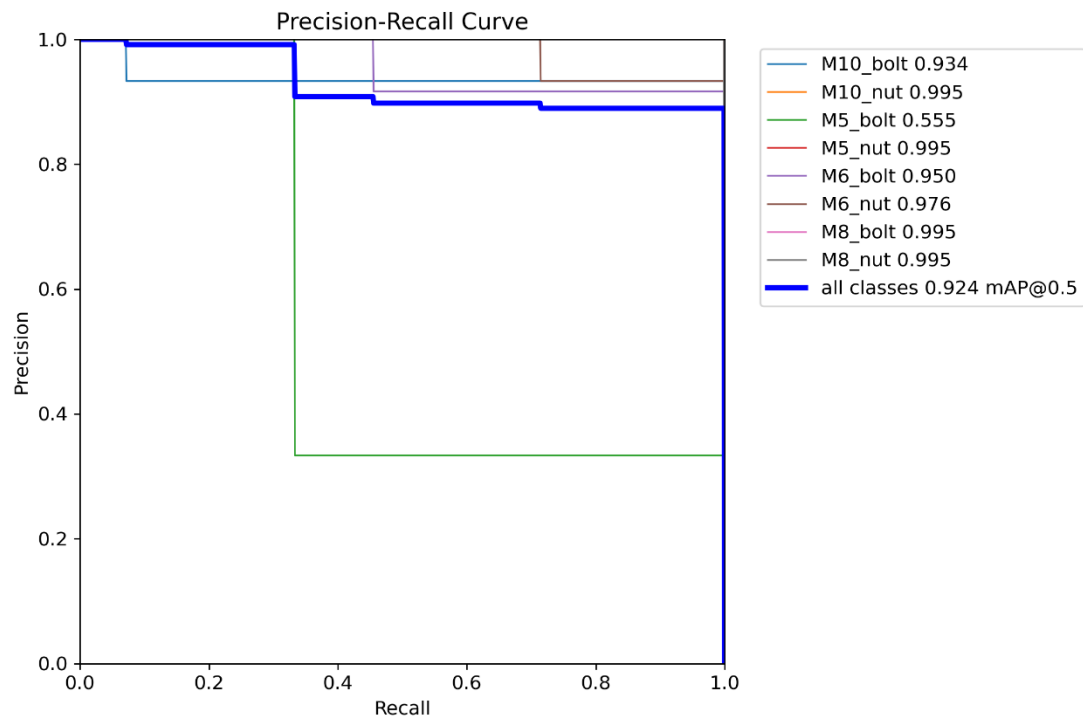


Figure 32: Precision–Recall Curve

Model performance can be measured graphically with the Precision-Recall (PR) curve. The Mean Average precision (mAP) at a threshold of .05 can be used to measure how well a model performs at detecting objects accurately; for example at 0.924 this would suggest the model has fairly marked performance when trying to detect object accurately.

The majority of classes were found to achieve high precision (or true positives over total predicted positives) with acceptable recall, indicating the model can reliably detect objects without generating a significant number of false positives. That said, one class (M5 bolts) did not demonstrate as high an object detection ability as that of larger component classes. Considering that the small component had fewer defining features, we would naturally expect poorer detection accuracy. Overall though, all curves do indicate stable and strong object detection performance.

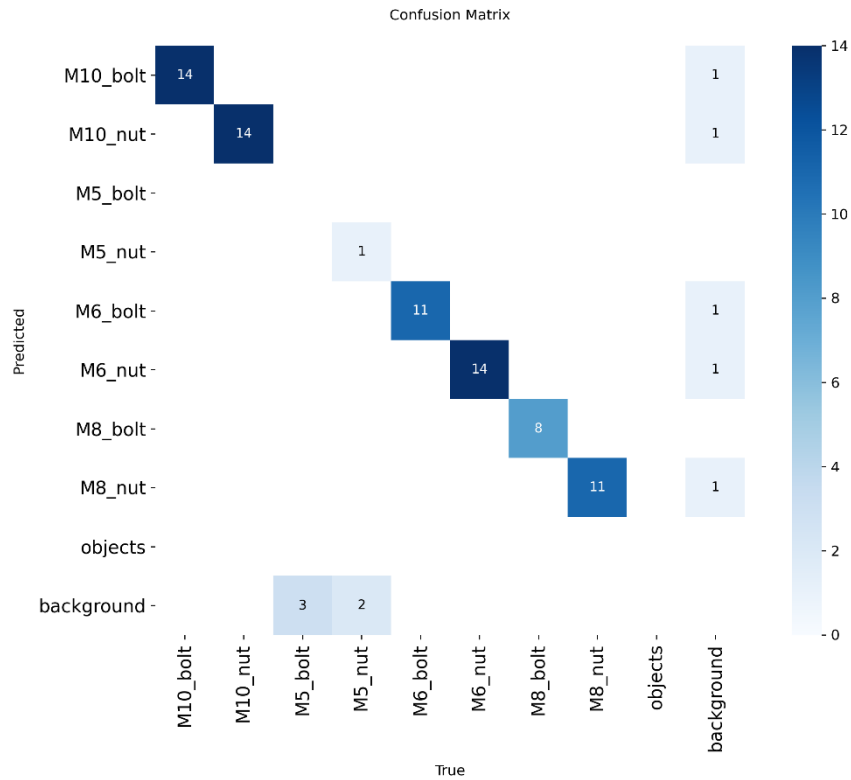


Figure 33: Confusion Matrix

The confusion matrix depicting predictions made by the model shown in Figure 33 primarily contains values in the diagonal area, indicating that objects were classified accurately. Classes M6 and M10 are classified with a fair degree of accuracy and confusion minimally exists amongst them. Misclassification rates have been slight, especially with components that are similar in appearance, such as nuts of varying sizes. A number of background detections were also observed, but this can occur in virtually any real-word dataset as well. However, these misclassifications do not significantly impact overall performance, as evidenced by the confusion matrix indicating that the model is trained sufficiently and can accurately distinguish between classes.



Figure 34: YOLOv8 Detection Output for Nut and Bolt Classification

YOLOv8 detects both a nut and a bolt at the same time in real time according to what Figure 34 shows. The confidence score of about 0.9 suggests that the prediction is quite reliable for both the M10 bolt and the M10 nut. The bounding boxes drawn around each object demonstrate that they were successfully identified, localized, classified, and captured accurately.

This also shows us that the model is capable of differentiating between a nut and a bolt (or two types of nuts) even though they may be next to each other. In addition, the consistent labels and confidence score of 0.9 suggest that the model has learned to identify distinguishing characteristics and attributes of the various classes very well.

As a practical implication of this success, it shows that this system can deliver real-time detection of objects in the industrial environment, which is a critical requirement for any type of automated sorting or traceability application. The other thing supported by the clear detection outputs is that the system will be able to connect with ERP type systems to enable the subsequent labeling and tracking of products.

Overall Analysis

The YOLOv8 model is performing very strongly during both training metric assessment and real-time detection. Reasonable confidence for each of the assessed objects, paired with excellent bounding boxes, illustrate how well the model can detect mechanical parts such as nuts and bolts.

The automated skill sets used to identify mechanical parts have proven accurate, maintaining the required balance of speed and accuracy needed when deploying on an embedded platform, such as Raspberry Pi, to achieve smooth automation of the manufacturing process.

The overall results of the YOLOv8 model have established it as an accurate, quick, and reliable method for use in real-world operations across all areas of industry, particularly in the applications of automated inspection, sorting, and traceability of manufactured products.

6.3 FABRICATION OF BEARING CLAMP AND MOTOR MOUNT



Figure 36: Laser-Cut Clamp Holder (Acrylic Plate)

The clamp holder plate was created by laser cutting an acrylic sheet, and is the primary structure to support and create a good alignment for the bearing assembly. The top of the plate has a large round hole, where the bearing or shaft will fit through, and has multiple smaller holes below the large hole for mounting and fastening purposes.

Using laser cutting provides high precision in the dimensions and smooth edge finishes, which are crucial for proper fit and alignment. Using multiple holes provides flexibility during assembly to allow proper mounting to the frame and integration with other components.



Figure 37: 3D Printed Motor Mount (Top View)

The three-dimensional (3D) model of the motor mount seen in Figure 37 is the top view of a three-dimensional printed motor mount that holds the motor and keeps it in alignment for optimal operation as part of the entire conveyor system. The central part of the motor mount is a semi-cylindrical shape that will provide the motor with a snug fit where it will remain stable.

To either side of the motor mount, there are flat plates, which are mounted to the frame; these plates contain multiple mounting holes to facilitate the ease of attaching the motor mount to the frame of the conveyor system. These holes are placed in such a way that provides the user with flexibility in placement of the motor and assists in achieving precise alignment of the motor with the pulley drive system for the conveyor. Keyword: Proper alignment of the motor drive components is critical to the proper operation of the conveyor, as well as to prevent any power loss resulting from a misalignment.

The production of the motor mount using 3D printing allows for a custom fit to the motor; this reduces the complexity of manufacturing the motor mount because of the flexibility of using a 3D printer. The design of the motor mount combines strength, stability, and adaptability; therefore, it would be an excellent design for the design of a prototype for the conveyor application.



Figure 38: 3D Printed Motor Mount (Isometric View)

The three-dimensional (3D) model of the motor mount seen in Figure 37 is the top view of a three-dimensional printed motor mount that holds the motor and keeps it in alignment for optimal operation as part of the entire conveyor system. The central part of the motor mount is a semi-cylindrical shape that will provide the motor with a snug fit where it will remain stable. To either side of the motor mount, there are flat plates, which are mounted to the frame; these plates contain multiple mounting holes to facilitate the ease of attaching the motor mount to the frame of the conveyor system. These holes are placed in such a way that provides the user with flexibility in placement of the motor and assists in achieving precise alignment of the motor with the pulley drive system for the conveyor. Keyword: Proper alignment of the motor drive components is critical to the proper operation of the conveyor, as well as to prevent any power loss resulting from a misalignment.

The production of the motor mount using 3D printing allows for a custom fit to the motor; this reduces the complexity of manufacturing the motor mount because of the flexibility of using a 3D printer. The design of the motor mount combines strength,

stability, and adaptability; therefore, it would be an excellent design for the design of a prototype for the conveyor application.

6.3 FULLY ASSEMBLY

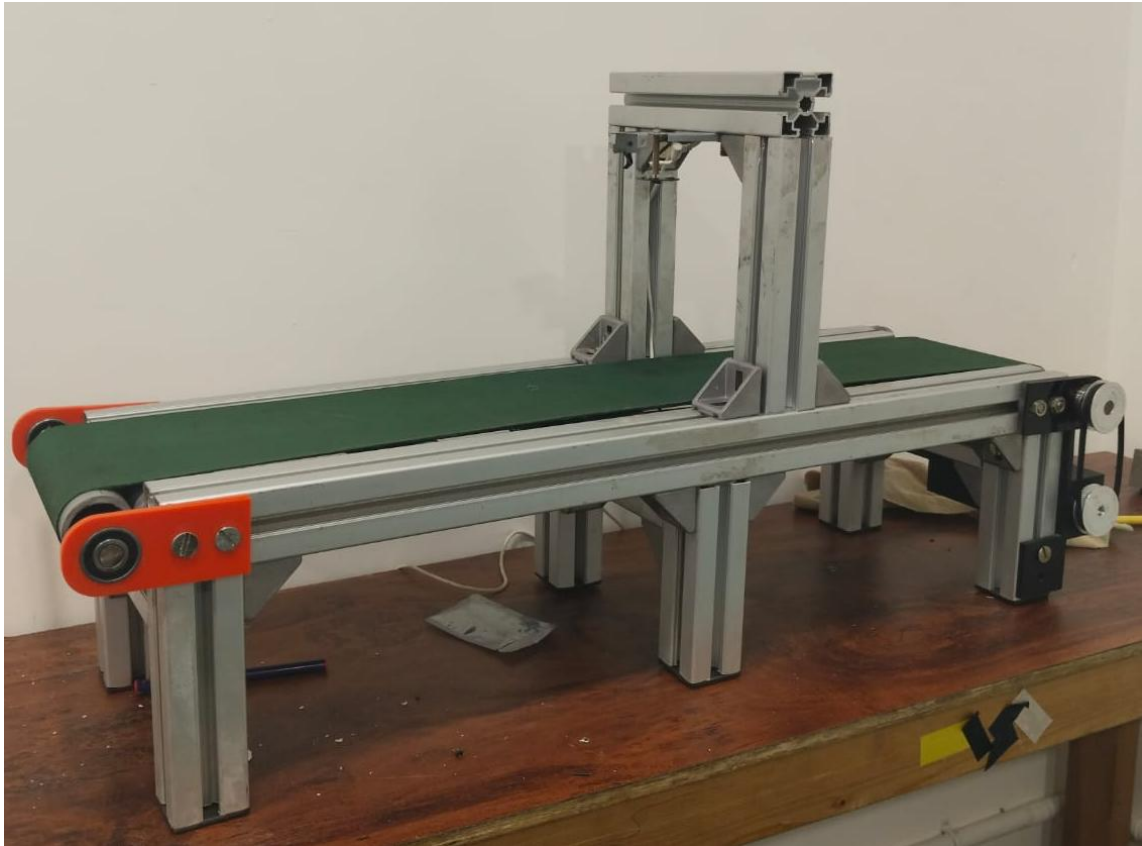


Figure 39: Complete Conveyor System Setup

Figure 39 illustrates the completed APTDS system with the mechanical, electronic and AI components combined as an operational unit in one mold. In addition, the aluminum profile frame structure is both robust and stable but also lightweight and can easily be modified. The conveyor belt is correctly positioned on the rollers, and the motor uses a timing belt for smooth, continuous operation. The system has also performed reliably during testing without vibration or misalignment. Therefore, it is an indication that a solid mechanical design was used during assembly of the system. The compactness of the APTDS makes it a suitable laboratory setup, yet still demonstrates the principles of operation found in an actual industrial application.



Figure 40: Conveyor Belt Operation View

Figure 40 provides a close-up view of the conveyor belt and roller mechanism in operation. The belt has the correct amount of tension in order to travel consistently over all of the rollers and does not slip or break its normal working path. This indicates the correct material was chosen when making the belt and that the pulleys have been aligned correctly.. Stability is very important for the detection of products because if products moved around the conveyor while travelling to their final destination, it may lead to improper detection. The shaft and bearing arrangement also aid in the reduction of friction, which increases the efficiency of the power transmission from the motor to the conveyor.



Figure 41: Detection Zone with Camera Mount

The area where the camera will be mounted above the conveyor via a rigid aluminum frame is illustrated in Figure 41. This arrangement allows for the camera to view clearly each part moving on the belt. The fixed mounting height and angle also help guarantee that the camera's view of the object will remain constant, which helps to increase the accuracy of the YOLOv8 model. The camera was able to capture clear images of nuts and bolts at various lighting conditions, which verifies that the camera and conveyor system will function appropriately for real-time object identification. The link between the physical aspects of the system (i.e., the conveyor, sensors and camera) and the AI processing is primarily through this section of the project.



Figure 42: Sensor and Camera Integration

Figure 42 illustrates the additional details to be integrated into the conveyor system for sensor/camera integration. All components are securely affixed to avoid any movement during operation. The sensor detects whether an object is located on the conveyor, while the camera provides an image for processing. Therefore, coordination between the sensor and camera allows for immediate action to be taken when necessary, such as stopping the conveyor for inspection or triggering detection. All wiring and placement throughout the entire system are simplified and organized; thus, making maintenance and troubleshooting of the system straightforward. Overall Analysis

The test results indicate the system has consistently performed as a fully automated system. The conveyor system delivered consistent speed with very little mechanical

loss, and the detection system detected parts consistently, with photographs captured clearly and extending to the accurate identification of components. The sensors, camera, and processing unit are integrated together, which produced good alignment between mechanical movement of the conveyor and pollution events.

Overall, the APTDS system has successfully automated the detection and tracking of parts, which reduces the amount of manual labour required; increases the accuracy of the information collected; substantially decreases the potential for human error; and continuously monitors the amount of component being produced. The findings demonstrate that the APTDS is both viable and scalable for use in the real-world and can be improved and customized for larger-scale enterprises. The successful completion of the project has created a reliable, efficient and comprehensive solution that integrates mechanical design with artificial intelligence and Enterprise Resource Planning (ERP).

CHAPTER 7

BILL OF MATERIAL

Table 3: Bill of materials

S.No	Component Name	Quantity	Unit Cost (₹)	Total Cost (₹)
1	Raspberry Pi 5	1	8000	8000
2	Pi Camera Module	1	1500	1500
3	DC Motor (12V)	1	800	800
4	Motor Driver (L298N)	1	300	300
5	Timing Belt	1	800	800
6	Pulleys (44mm dia)	2	250	500
7	Bearings (32mm)	16	40	640
8	Acrylic Sheets (Laser Cut)	12	50	600
9	3D Printed Parts	1 set	800	800
10	Aluminum Extrusion (800 mm)	3 sets	1500	4500
11	L-Clamps	14	80	1120
12	End Closers	8	20	160
13	Fasteners (Bolts, Nuts, etc.)	1 set	300	300
14	Power Supply (12V Adapter)	1	600	600
15	Wiring and Connectors	1 set	300	300
16	Conveyor Belt (Fabric)	1	800	800

Total Estimated Cost: ₹21,720

Description

This Bill of Materials has been updated to include the actual parts used to create the APTDS device. The principal portion of the costs has to do with major components such as: Raspberry Pi; aluminum frame and the camera module; these are required for proper processing and system reliability.

Mechanical components (bearings; pulleys; timing belt; etc.) allow for the movement of the conveyor smoothly. The structural components (aluminum extrusions and L-clamps) provide an adequate framework to support the system and help maintain its durability. The 3D-printed and laser-cut components help to eliminate complexity in producing the device and allow for customization with relative ease.

The costs associated with this system are very reasonable based upon its capabilities, how much automation the unit can perform, and its ability to perform in real-time. Therefore, this system is appropriate for use in both academic projects as well as ... using this equipment in small-scale industrial applications.

CHAPTER 8

Conclusion

The Automated Part Traceability and Data Synchronization project demonstrated how the combination of advanced technologies such as computer vision, embedded systems, and mechanical automation can produce a more efficient solution at an industrial level. The APTDS system was created to automate the identification, confirmation, and tracking of component parts (e.g., nuts and bolts) as they travel along a conveyor belt. Through the integration of a Raspberry Pi 5 with a camera module and the YOLOv8 model, the APTDS system is capable of accurately detecting and identifying around multiple objects moving down the conveyor at a much faster rate than could be accomplished manually.

One of the main accomplishments of the APTDS project is the establishment of a structured barcode system. Each may object created will include a unique barcode that encodes the item's product characteristics (such as standard, batch number, and time/date created). As such, the APTDS system has developed a means to trace parts back to their origin from start to finish. Finally, as the APTDS project integrates the barcodes into ERP system, the overall efficiency of tracking parts through the APTDS system will improve.

Related to the mechanical aspects of the APTDS project, the construction of the conveyor system utilized aluminium extrusion, timing belts, pulleys and bearings to create a stable and smooth conveyor belt. In addition, the use of laser cut acrylic, coupled with approximately 35 3D-printable components allowed for flexibility in design and reduced the cost of materials needed.

The project has achieved its goal of building a smart, automated and more dependable part identification and tracking system. It provided a clear demonstration of how similar systems could be used in manufacturing environments to improve productivity and maintain quality control.

Discussion

The results yielded from this system have demonstrated both the advantages and disadvantages of this type of system. YOLOv8 is the most appropriate model for

application to a real-time conveyor system. Even though other models (Faster R-CNN) produced slightly better accuracy than YOLOv8, they had longer processing times which resulted in overall in-feasibility for use on an embedded platform (Raspberry-Pi). This exemplifies the engineering principle of trade-offs; speed and efficiency can be just as important as accuracy in terms of applicable, real-time applications.

The charts and metrics contained in this report demonstrate that the best Object Recognition results were achieved by the majority of the classes explored; especially true for object classes consisting of objects with medium to large size (M8 and M10 nuts and bolts for example). However, the small item classes (e.g., M5 size) did not yield as good results. The reduced recognition accuracy for these smaller items is due to the decreased size, decreased distinct features, and increased lighting sensitivity of smaller items.

The design of modular hardware has proven to be highly successful in this proposed solution. The extrusion of aluminum frames allowed for the ease of assembly and alignment via L-clamps and end connectors while providing the needed structural rigidity of the system. The use of laser-cut and 3D printed parts helped to reduce the complexity and costs associated with the manufacturing of the prototype while providing the functionality of the prototype. There were minor challenges identified during the testing of the prototype, including: belt alignment; vibration; and mechanical tolerances that are typical of prototype systems. All of these areas can be improved through the use of proper manufacturing precision and industrial-grade components.

The proposed system provides an ERP integration that gives real-time access to the monitoring and data logging of production-related information. This enhancement allows the proposed system to change from a basic detection system to a full-fledged integrated smart manufacturing system that allows an operator complete visibility into production efficiencies, errors, and maintaining accurate data records.

Finally, there are a variety of upgrades that could be completed to the current prototype in order to improve it in the future. Hardware acceleration techniques (Tensors) can be used to maximize the performance of the processing of the system. The replacement of the current camera that will detect products and the appropriate control of the lighting of the environment will enhance the ability to detect smaller products. Finally, the use

of additional sensors (ultrasonic or proximity) that provide more feedback on the use of the motors will further enhance the reliability of the system.

Overall, the APTDS system has successfully reached its design specifications while also showing how AI, IoT, and mechanical systems can work together in industrial automation. The system serves as an excellent basis for additional developments/extensions, and it can be modified or expanded on to accommodate more complicated forms of manufacturing.

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